MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3810 RGENT- CAUST. SODA & CYAN. MIX. STN.
DESIGN OF FOUNDATION

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4
2. EXECUTIVE SUMMARY 4
3. SCOPE 4
4. DESIGN REQUIREMENTS 5
   - 4.1 Design Loads 5
5. CONCRETE FOUNDATIONS 5
   - 5.1 Design 5
   - 5.2 Materials 5
   - 5.3 Minimum Concrete Cover 6
   - 5.4 Allowable Soil Bearing Capacity 6
6. DESIGN CALCULATION FOR FOUNDATION 7

References 18

ATTACHMENT 1 INPUT DRAWINGS 19

ATTACHMENT 2 STAAD INPUT - FOUNDATION F1 22
ATTACHMENT 3 STAAD INPUT - FOUNDATION F2 27
ATTACHMENT 4 STAAD OUTPUT - FOUNDATION F1 31
ATTACHMENT 5 STAAD OUTPUT - FOUNDATION F2 42

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RGENT.	:	Reagents
CAUST.	:	Caustic
CYAN.	:	Cyanide
MIX.	:	Mixing
STN.	:	Station

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1. INTENT

This document is to prepare design of foundation for **Caustic & cyanide storage & mixing station (3810)**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Doc. No. MD-216-3810-EG-ST-GAD-2344 attached in Attachment 1. Loads are also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.5 m below EGL. (Refer Table 5 - BH-15 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-Z	Along-X
2	0.50m	0.50m

Size of fdn. raft:	3.75m	3.75m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

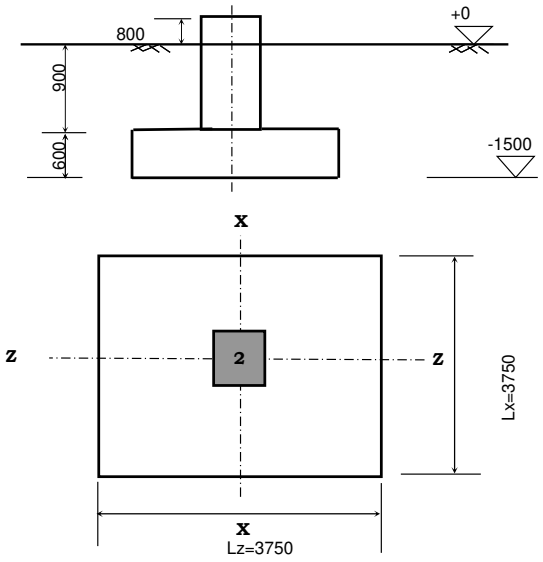
Weight of raft = (3.75*3.75*0.6*24) = 202.50KN

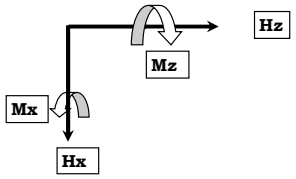
Weight of soil = ((3.75*3.75)-(0.5*0.5))*0.9*20 = 248.63KN

Total = 451.13KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL + CL	33.18	-10.10	23.23	0.00	0.00
202 DL+CL+L+TL1	79.88	-44.10	101.43	0.00	0.00
203 DL+CL+L+TL2	79.88	-44.90	103.27	0.00	0.00
204 DL+CL+0.75L+0.75TL1	68.20	-35.60	81.88	0.00	0.00
205 DL+CL+0.75L+0.75TL2	68.20	-36.20	83.26	0.00	0.00
206 DL+CL+WL1	-95.55	59.60	-137.08	-40.10	-92.23
207 DL+CL+WL2	-70.45	26.90	-61.87	-40.10	-92.23
208 DL+CL+WL3	-56.35	40.30	-92.69	-40.10	-92.23
209 DL+CL+WL4	-31.25	7.50	-17.25	-40.10	-92.23
210 DL+CL+WL5	-124.15	28.60	-65.78	-40.10	-92.23
211 DL+CL+WL6	-107.85	31.60	-72.68	-40.10	-92.23
212 DL+CL+0.75WL1+0.75L	6.69	-9.43	21.68	-30.08	-69.17
213 DL+CL+0.75WL2+0.75L	25.51	-33.95	78.09	-30.08	-69.17
214 DL+CL+0.75WL3+0.75L	36.09	-23.90	54.97	-30.08	-69.17
215 DL+CL+0.75WL4+0.75L	54.91	-48.50	111.55	-30.08	-69.17
216 DL+CL+0.75WL5+0.75L	-14.77	-32.68	75.15	-30.08	-69.17
217 DL+CL+0.75WL67+0.75L	-2.54	-30.43	69.98	-30.08	-69.17
218 0.6DL+CL+WL1	-107.98	63.60	-146.28	-40.10	-92.23
219 0.6DL+CL+WL2	-82.88	30.90	-71.07	-40.10	-92.23
220 0.6DL+CL+WL3	-68.78	44.30	-101.89	-40.10	-92.23
221 0.6DL+CL+WL4	-43.68	11.50	-26.45	-40.10	-92.23
222 0.6DL+CL+WL5	-136.58	32.60	-74.98	-40.10	-92.23
223 0.6DL+CL+WL6	-120.28	35.60	-81.88	-40.10	-92.23
224 0.6DL+CL+0.7SL1	18.72	-6.45	14.84	-1.96	-4.51
225 0.6DL+CL+0.7SL2	19.00	-5.75	13.23	-1.96	-4.51

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL + CL	484.3	0.00	0.0	-10.1	23.2	39.1	NA	19.2	NA
202 DL+CL+L+TL1	531.0	0.00	0.0	-44.1	101.4	9.8	NA	4.8	NA
203 DL+CL+L+TL2	531.0	0.00	0.0	-44.9	103.3	9.6	NA	4.7	NA
204 DL+CL+0.75L+0.75TL1	519.3	0.00	0.0	-35.6	81.9	11.9	NA	5.8	NA
205 DL+CL+0.75L+0.75TL2	519.3	0.00	0.0	-36.2	83.3	11.7	NA	5.7	NA
206 DL+CL+WL1	355.6	-40.10	-92.2	59.6	-137.1	4.9	7.2	2.4	3.5
207 DL+CL+WL2	380.7	-40.10	-92.2	26.9	-61.9	11.5	7.7	5.7	3.8
208 DL+CL+WL3	394.8	-40.10	-92.2	40.3	-92.7	8.0	8.0	3.9	3.9
209 DL+CL+WL4	419.9	-40.10	-92.2	7.5	-17.3	45.6	8.5	22.4	4.2
210 DL+CL+WL5	327.0	-40.10	-92.2	28.6	-65.8	9.3	6.6	4.6	3.3
211 DL+CL+WL6	343.3	-40.10	-92.2	31.6	-72.7	8.9	7.0	4.3	3.4
212 DL+CL+0.75WL1+0.75L	457.8	-30.08	-69.2	-9.4	21.7	39.6	12.4	19.4	6.1
213 DL+CL+0.75WL2+0.75L	476.6	-30.08	-69.2	-34.0	78.1	11.4	12.9	5.6	6.3
214 DL+CL+0.75WL3+0.75L	487.2	-30.08	-69.2	-23.9	55.0	16.6	13.2	8.2	6.5
215 DL+CL+0.75WL4+0.75L	506.0	-30.08	-69.2	-48.5	111.6	8.5	13.7	4.2	6.7
216 DL+CL+0.75WL5+0.75L	436.4	-30.08	-69.2	-32.7	75.2	10.9	11.8	5.3	5.8
217 DL+CL+0.75WL67+0.75L	448.6	-30.08	-69.2	-30.4	70.0	12.0	12.2	5.9	6.0
218 0.6DL+CL+WL1	343.2	-40.10	-92.2	63.6	-146.3	4.4	7.0	2.2	3.4
219 0.6DL+CL+WL2	368.3	-40.10	-92.2	30.9	-71.1	9.7	7.5	4.8	3.7
220 0.6DL+CL+WL3	382.4	-40.10	-92.2	44.3	-101.9	7.0	7.8	3.5	3.8
221 0.6DL+CL+WL4	407.5	-40.10	-92.2	11.5	-26.5	28.9	8.3	14.2	4.1
222 0.6DL+CL+WL5	314.6	-40.10	-92.2	32.6	-75.0	7.9	6.4	3.9	3.1
223 0.6DL+CL+WL6	330.9	-40.10	-92.2	35.6	-81.9	7.6	6.7	3.7	3.3
224 0.6DL+CL+0.7SL1	469.8	-1.96	-4.5	-6.5	14.8	59.4	195.4	29.1	95.9
225 0.6DL+CL+0.7SL2	470.1	-1.96	-4.5	-5.8	13.2	66.7	195.5	32.7	95.9

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	56.33	0
Min Px	2	201	0	56.33	0
Max Py	587	215	0	74.35	0
Min Py	570	218	0	1.90	0
Max Pz	2	201	0	56.33	0
Min Pz	2	201	0	56.33	0

Gross Maximum Base pressure, p_{max} = 74.4 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.9 m
Unit weight of soil = 20 KN/m^3

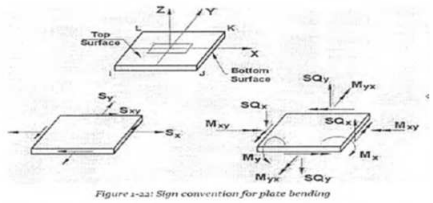
Gross Safe Bearing Capacity = 368 KN/m^2
> 74.4 KN/m^2
Hence OK

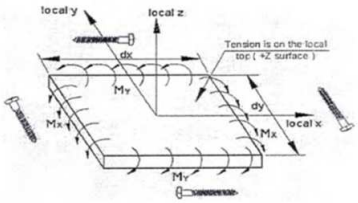
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Design of foundation raft F1





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.245	-0.087	0	0	0	-64.37	-25.962	8.059	-72.425	-34.021	103:1.2DL+1.2CL+1.2TL2+1.6L
Min Qx	427	-0.425	-0.055	0	0	0	104.07	36.125	20.851	124.924	56.976	113:0.9DL+0.9CL+1.6WL1
Max Qy	333	-0.32	0.21	0	0	0	91.308	43.45	-3.491	94.799	46.941	113:0.9DL+0.9CL+1.6WL1
Min Qy	416	0.034	-0.355	0	0	0	28.87	77.837	20.294	49.164	98.131	113:0.9DL+0.9CL+1.6WL1
Max Sx	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sx	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Sy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Sxy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sxy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Mx	427	-0.425	-0.055	0	0	0	104.07	36.125	20.851	124.924	56.976	113:0.9DL+0.9CL+1.6WL1
Min Mx	327	-0.22	-0.014	0	0	0	-69.16	17.558	-26.361	-95.52	43.919	113:0.9DL+0.9CL+1.6WL1
Max My	416	0.034	-0.355	0	0	0	28.87	77.837	20.294	49.164	98.131	113:0.9DL+0.9CL+1.6WL1
Min My	345	0	-0.165	0	0	0	6.929	-47.519	-17.658	24.587	-65.177	107:1.2DL+1.2CL+1.6WL3+0.5L
Max Mxy	417	-0.283	-0.323	0	0	0	71.875	64.369	28.861	100.736	93.23	113:0.9DL+0.9CL+1.6WL1
Min Mxy	344	-0.157	-0.196	0	0	0	-37.73	-30.083	-40.1	-77.795	-70.145	113:0.9DL+0.9CL+1.6WL1

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	337	0.084	-0.016	0	0	0	-16.61	-20.301	-0.787	-17.394	-21.088	103:1.2DL+1.2CL+1.2TL2+1.6L
Min Qx	429	-0.138	0.005	0	0	0	25.388	33.366	5.259	30.647	38.625	113:0.9DL+0.9CL+1.6WL1
Max Qy	349	-0.073	0.074	0	0	0	22.464	18.025	8.088	30.552	26.113	113:0.9DL+0.9CL+1.6WL1
Min Qy	397	-0.019	-0.117	0	0	0	32.094	17.452	4.754	36.848	22.206	113:0.9DL+0.9CL+1.6WL1
Max Sx	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sx	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Sy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Sxy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Min Sxy	588	0	0	0	0	0	-0.026	-0.018	-0.004	-0.03	-0.022	101:1.4DL+1.4CL
Max Mx	397	-0.019	-0.117	0	0	0	32.094	17.452	4.754	36.848	22.206	113:0.9DL+0.9CL+1.6WL1
Min Mx	337	0.084	-0.016	0	0	0	-16.61	-20.301	-0.787	-17.394	-21.088	103:1.2DL+1.2CL+1.2TL2+1.6L
Max My	429	-0.138	0.005	0	0	0	25.388	33.366	5.259	30.647	38.625	113:0.9DL+0.9CL+1.6WL1
Min My	337	0.084	-0.016	0	0	0	-16.61	-20.301	-0.787	-17.394	-21.088	103:1.2DL+1.2CL+1.2TL2+1.6L
Max Mxy	395	0.064	-0.078	0	0	0	21.422	17.339	11.546	32.968	28.885	113:0.9DL+0.9CL+1.6WL1
Min Mxy	365	0.04	-0.032	0	0	0	2.158	-8.081	-11.559	13.717	-19.64	113:0.9DL+0.9CL+1.6WL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

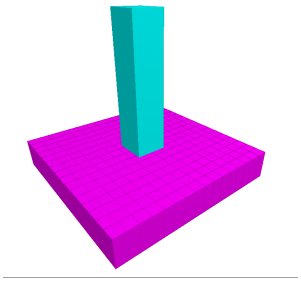
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
113:0.9D	124.92	top face	50	542	124.92	0.57	0.0014	738.8	975.6	16	206.17	200	1005.8	14.2	17.73	0.089	tension control
113:0.9D	-95.52	bottom face	50	542	95.52	0.43	0.0010	563.6		16	206.17	200	1005.8	14.2	17.73	0.089	tension control
113:0.9D	100.74	top face	50	542	100.74	0.46	0.0011	594.6		16	206.17	200	1005.8	14.2	17.73	0.089	tension control
113:0.9D	-77.80	bottom face	50	542	77.795	0.35	0.0008	458.4		16	206.17	200	1005.8	14.2	17.73	0.089	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.084	1	50.4	17.394	0.002	0.542	539.895	530.141	929.889	530.14	1985	> 50.4	ok	397.6	φVc>Vu	not required
	Min Qx	0.138	1	82.8	30.647			538.081			530.14	1985	> 82.8	ok	397.6	φVc>Vu	not required

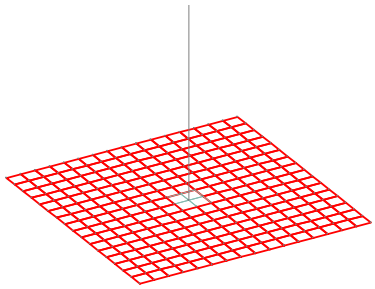
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
113:0.9D	98.13	top face	50	542	98.131	0.45	0.0011	579.1	975.6	16	206.17	200	1005.8	14.2	17.73	0.089	tension control
107:1.2D	-65.18	bottom face	50	542	65.177	0.30	0.0007	383.7		16	206.17	200	1005.8	14.2	17.73	0.089	tension control
113:0.9D	93.23	top face	50	542	93.23	0.42	0.0010	550.0		16	206.17	200	1005.8	14.2	17.73	0.089	tension control
113:0.9D	-70.15	bottom face	50	542	70.145	0.32	0.0008	413.1		16	206.17	200	1005.8	14.2	17.73	0.089	tension control

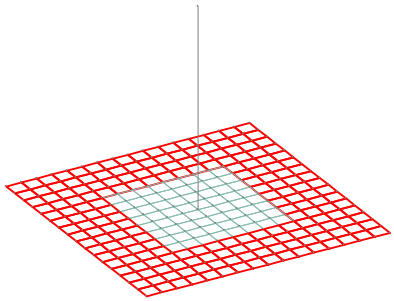
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.074	1	44.4	26.113	0.002	0.542	528.8	530.141	fc'929.9	528.8	555	> 44.4	ok	396.6	φVc>Vu	not required
	Min Qy	0.117	1	70.2	22.206			542.34			530.14	556	> 70.2	ok	397.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16φT**	**200**	c/c
At Top	**16φT**	**200**	c/c
Along Z Direction :			
At Bottom	**16φT**	**200**	c/c
At Top	**16φT**	**200**	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-Z	Along-X
2	0.50m	0.50m

Size of fdn. raft:	2.50m	2.50m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.80	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

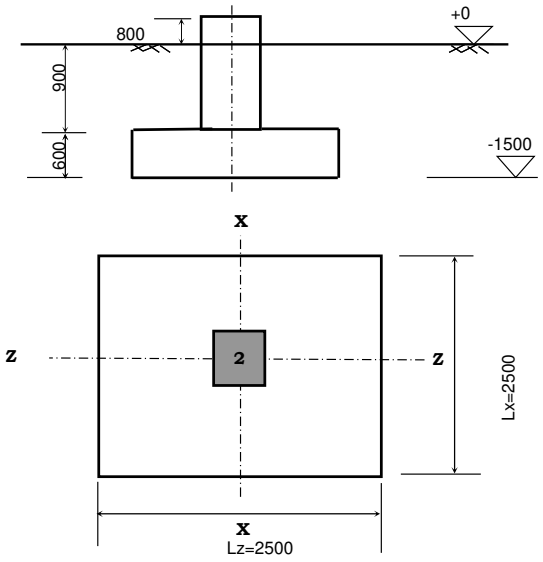
Weight of raft = (2.5*2.5*0.6*24) = 90.00KN

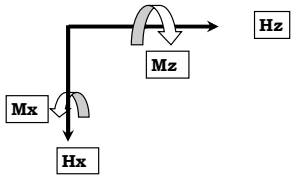
Weight of soil = ((2.5*2.5)-(0.5*0.5))*0.9*20 = 108.00KN

Total = 198.00KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL + CL	20.48	0.00	0.00	0.00	0.00
202 DL+CL+L+TL1	30.98	0.00	0.00	0.00	0.00
203 DL+CL+L+TL2	36.38	0.00	0.00	0.00	0.00
204 DL+CL+0.75L+0.75TL1	28.35	0.00	0.00	0.00	0.00
205 DL+CL+0.75L+0.75TL2	32.40	0.00	0.00	0.00	0.00
206 DL+CL+WL1	-10.83	0.00	0.00	-22.30	-51.29
207 DL+CL+WL2	-8.13	0.00	0.00	-24.60	-56.58
208 DL+CL+WL3	-1.03	0.00	0.00	-24.60	-56.58
209 DL+CL+WL4	1.68	0.00	0.00	-22.30	-51.29
210 DL+CL+WL5	-6.33	0.00	0.00	-22.30	-51.29
211 DL+CL+WL6	7.88	0.00	0.00	-24.60	-56.58
212 DL+CL+0.75WL1+0.75L	16.80	0.00	0.00	-16.73	-38.47
213 DL+CL+0.75WL2+0.75L	18.83	0.00	0.00	-18.45	-42.44
214 DL+CL+0.75WL3+0.75L	24.15	0.00	0.00	-18.45	-42.44
215 DL+CL+0.75WL4+0.75L	26.18	0.00	0.00	-16.73	-38.47
216 DL+CL+0.75WL5+0.75L	20.18	0.00	0.00	-16.73	-38.47
217 DL+CL+0.75WL67+0.75L	30.83	0.00	0.00	-18.45	-42.44
218 0.6DL+CL+WL1	-18.94	0.00	0.00	-22.30	-51.29
219 0.6DL+CL+WL2	-16.24	0.00	0.00	-24.60	-56.58
220 0.6DL+CL+WL3	-9.14	0.00	0.00	-24.60	-56.58
221 0.6DL+CL+WL4	-6.44	0.00	0.00	-22.30	-51.29
222 0.6DL+CL+WL5	-14.44	0.00	0.00	-22.30	-51.29
223 0.6DL+CL+WL6	-0.24	0.00	0.00	-24.60	-56.58
224 0.6DL+CL+0.7SL1	12.72	0.00	0.00	0.00	0.00
225 0.6DL+CL+0.7SL2	12.02	0.00	0.00	0.00	0.00

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 DL + CL	218.5	0.00	0.0	0.0	0.0	NA	NA	NA	NA
202 DL+CL+L+TL1	229.0	0.00	0.0	0.0	0.0	NA	NA	NA	NA
203 DL+CL+L+TL2	234.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
204 DL+CL+0.75L+0.75TL1	226.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
205 DL+CL+0.75L+0.75TL2	230.4	0.00	0.0	0.0	0.0	NA	NA	NA	NA
206 DL+CL+WL1	187.2	-22.30	-51.3	0.0	0.0	NA	4.6	NA	3.4
207 DL+CL+WL2	189.9	-24.60	-56.6	0.0	0.0	NA	4.2	NA	3.1
208 DL+CL+WL3	197.0	-24.60	-56.6	0.0	0.0	NA	4.4	NA	3.2
209 DL+CL+WL4	199.7	-22.30	-51.3	0.0	0.0	NA	4.9	NA	3.6
210 DL+CL+WL5	191.7	-22.30	-51.3	0.0	0.0	NA	4.7	NA	3.4
211 DL+CL+WL6	205.9	-24.60	-56.6	0.0	0.0	NA	4.5	NA	3.3
212 DL+CL+0.75WL1+0.75L	214.8	-16.73	-38.5	0.0	0.0	NA	7.0	NA	5.1
213 DL+CL+0.75WL2+0.75L	216.8	-18.45	-42.4	0.0	0.0	NA	6.4	NA	4.7
214 DL+CL+0.75WL3+0.75L	222.2	-18.45	-42.4	0.0	0.0	NA	6.5	NA	4.8
215 DL+CL+0.75WL4+0.75L	224.2	-16.73	-38.5	0.0	0.0	NA	7.3	NA	5.4
216 DL+CL+0.75WL5+0.75L	218.2	-16.73	-38.5	0.0	0.0	NA	7.1	NA	5.2
217 DL+CL+0.75WL67+0.75L	228.8	-18.45	-42.4	0.0	0.0	NA	6.7	NA	5.0
218 0.6DL+CL+WL1	179.1	-22.30	-51.3	0.0	0.0	NA	4.4	NA	3.2
219 0.6DL+CL+WL2	181.8	-24.60	-56.6	0.0	0.0	NA	4.0	NA	3.0
220 0.6DL+CL+WL3	188.9	-24.60	-56.6	0.0	0.0	NA	4.2	NA	3.1
221 0.6DL+CL+WL4	191.6	-22.30	-51.3	0.0	0.0	NA	4.7	NA	3.4
222 0.6DL+CL+WL5	183.6	-22.30	-51.3	0.0	0.0	NA	4.5	NA	3.3
223 0.6DL+CL+WL6	197.8	-24.60	-56.6	0.0	0.0	NA	4.4	NA	3.2
224 0.6DL+CL+0.7SL1	210.7	0.00	0.0	0.0	0.0	NA	NA	NA	NA
225 0.6DL+CL+0.7SL2	210.0	0.00	0.0	0.0	0.0	NA	NA	NA	NA

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	56.31	0
Min Px	2	201	0	56.31	0
Max Py	357	211	0	75.11	0
Min Py	372	219	0	7.61	0
Max Pz	2	201	0	56.31	0
Min Pz	2	201	0	56.31	0

Gross Maximum Base pressure, p_{max} = 75.1 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.9 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 368 KN/m^2
> 75.1 KN/m^2
Hence OK

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Design of foundation raft F1

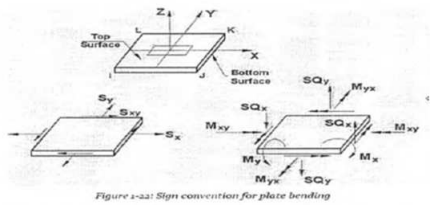
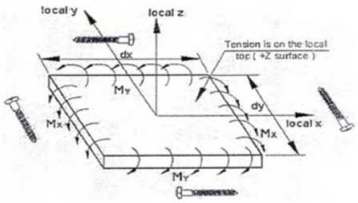


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	414	0.116	-0.063	0	0	0	8.846	16.016	-5.641	14.487	21.657	114:0.9DL+0.9CL+1.6WL2
Min Qx	417	-0.116	-0.063	0	0	0	8.846	16.016	5.641	14.487	21.657	114:0.9DL+0.9CL+1.6WL2
Max Qy	333	0.024	0.08	0	0	0	-1.137	5.332	5.149	-6.286	10.481	114:0.9DL+0.9CL+1.6WL2
Min Qy	416	-0.061	-0.161	0	0	0	10.066	28.545	3.768	13.834	32.313	114:0.9DL+0.9CL+1.6WL2
Max Sx	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sx	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Sy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Sxy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sxy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Mx	416	-0.061	-0.161	0	0	0	10.066	28.545	3.768	13.834	32.313	114:0.9DL+0.9CL+1.6WL2
Min Mx	356	0.019	-0.079	0	0	0	-8.303	-11.045	0.149	-8.452	-11.194	110:1.2DL+1.2CL+1.6WL6+0.5L
Max My	416	-0.061	-0.161	0	0	0	10.066	28.545	3.768	13.834	32.313	114:0.9DL+0.9CL+1.6WL2
Min My	346	0.06	-0.151	0	0	0	-8.09	-27.548	4.081	-12.171	-31.629	110:1.2DL+1.2CL+1.6WL6+0.5L
Max Mxy	347	0.094	-0.041	0	0	0	-5.112	-12.268	7.248	-12.36	-19.516	114:0.9DL+0.9CL+1.6WL2
Min Mxy	344	-0.094	-0.041	0	0	0	-5.112	-12.268	-7.2	-12.36	-19.516	114:0.9DL+0.9CL+1.6WL2

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	404	0.045	-0.053	0	0	0	6.134	8.966	-0.281	6.415	9.247	114:0.9DL+0.9CL+1.6WL2
Min Qx	407	-0.045	-0.053	0	0	0	6.134	8.966	0.281	6.415	9.247	114:0.9DL+0.9CL+1.6WL2
Max Qy	334	0.01	0.032	0	0	0	0.283	-1.588	5.24	5.523	-6.828	114:0.9DL+0.9CL+1.6WL2
Min Qy	406	-0.021	-0.084	0	0	0	9.685	11.447	0.006	9.691	11.453	114:0.9DL+0.9CL+1.6WL2
Max Sx	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sx	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Sy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Sxy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Min Sxy	432	-0.005	0.001	0	0	0	-0.203	-1.907	-0.152	-0.355	-2.059	101:1.4DL+1.4CL
Max Mx	406	-0.021	-0.084	0	0	0	9.685	11.447	0.006	9.691	11.453	114:0.9DL+0.9CL+1.6WL2
Min Mx	356	0.019	-0.079	0	0	0	-8.303	-11.045	0.149	-8.452	-11.194	110:1.2DL+1.2CL+1.6WL6+0.5L
Max My	406	-0.021	-0.084	0	0	0	9.685	11.447	0.006	9.691	11.453	114:0.9DL+0.9CL+1.6WL2
Min My	356	0.019	-0.079	0	0	0	-8.303	-11.045	0.149	-8.452	-11.194	110:1.2DL+1.2CL+1.6WL6+0.5L
Max Mxy	334	0.01	0.032	0	0	0	0.283	-1.588	5.24	5.523	-6.828	114:0.9DL+0.9CL+1.6WL2
Min Mxy	326	-0.01	0.032	0	0	0	0.283	-1.588	-5.24	5.523	-6.828	114:0.9DL+0.9CL+1.6WL2

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

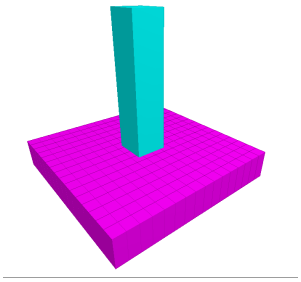
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
114:0.9D	13.83	top face	50	442	13.834	0.09	0.0002	99.5	795.6	16	252.82	200	1005.8	14.2	17.73	0.072	tension control
110:1.2D	-8.45	bottom face	50	442	8.452	0.06	0.0001	60.8		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
114:0.9D	-12.36	bottom face	50	442	12.36	0.08	0.0002	88.9		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
114:0.9D	-12.36	bottom face	50	442	12.36	0.08	0.0002	88.9		16	252.82	200	1005.8	14.2	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.045	1	22.5	6.415	0.002	0.442	444.893	435.484	758.323	435.48	1621	> 22.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.045	1	22.5	6.415			444.893			435.48	1621	> 22.5	ok	326.6	φVc>Vu	not required

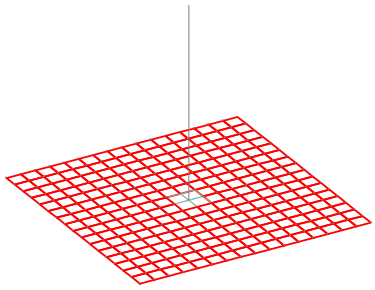
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
114:0.9D	32.31	top face	50	442	32.313	0.22	0.0005	233.0	795.6	16	252.82	200	1005.8	14.2	17.73	0.072	tension control
110:1.2D	-31.63	bottom face	50	442	31.629	0.22	0.0005	228.0		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
114:0.9D	-19.52	bottom face	50	442	19.516	0.13	0.0003	140.5		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
114:0.9D	-19.52	bottom face	50	442	19.516	0.13	0.0003	140.5		16	252.82	200	1005.8	14.2	17.73	0.072	tension control

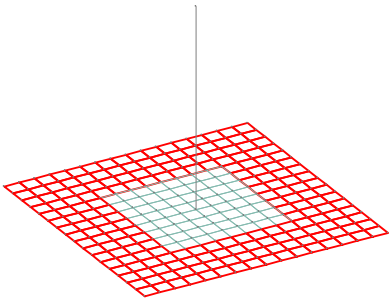
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.032	1	16	6.828	0.002	0.442	436.095	435.484	fc'758.3	435.48	456	> 16	ok	326.6	φVc>Vu	not required
	Min Qy	0.084	1	42	11.453			446.1			435.48	456	> 42	ok	326.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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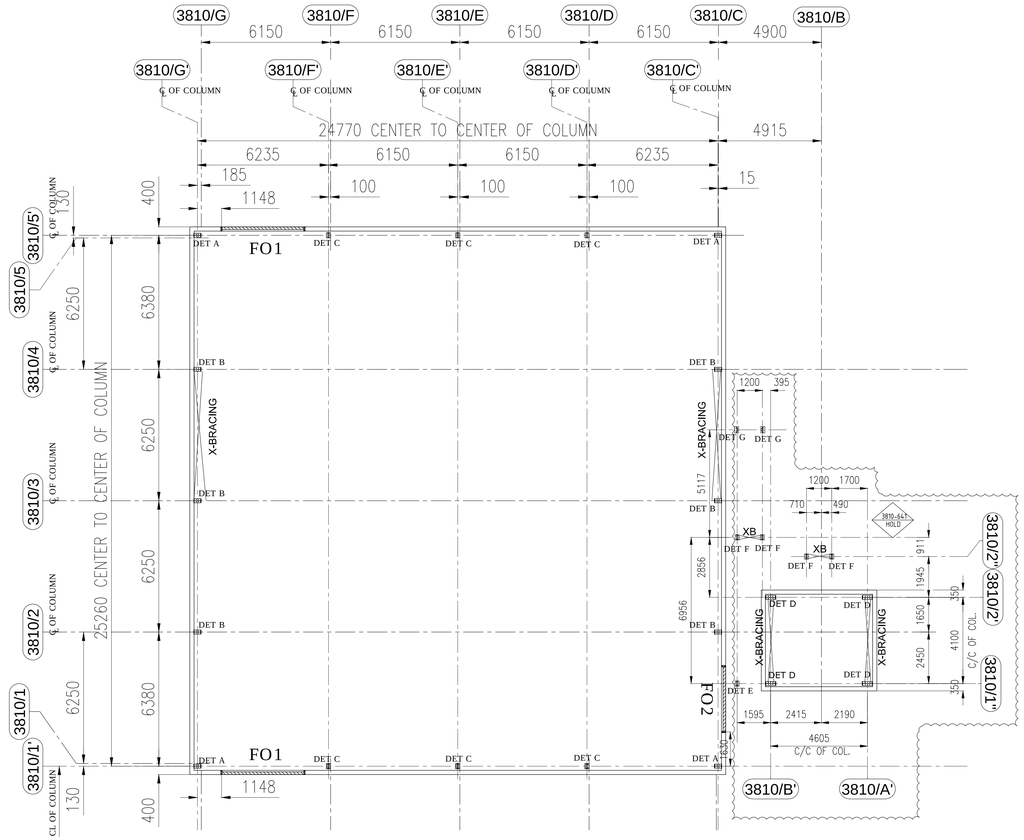
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ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER: MD-216-3810-EG-ST-GAD-2343 | REV: A01

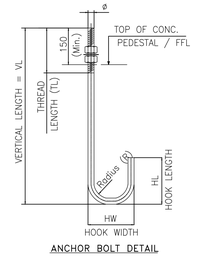


NOTE: ANCHOR BOLT SPECIFICATION (AS PER ASTM 1554, GR.36,50,105)

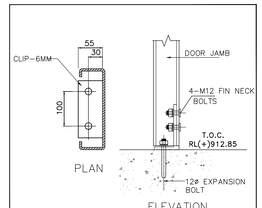
QTY.	DIA. OF BOLT=ø	DIA. OF HOLE	THREAD LENGTH	HOOK LENGTH	HOOK WIDTH	RADIUS	VERTICAL LENGTH (VL)	ANC.ROD LENGTH
--	ø	D	TL	HL	HW	R	VL	AL
0	M16	22	150	110	200	100	575	710
68	M20	26	150	110	200	100	575	710
24	M24	30	200	140	256	128	675	851
--	M30	36	250	170	320	160	975	1189
--	M36	42	250	210	384	192	1075	1338

REFERENCE NOTES :-

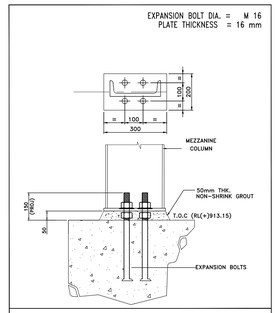
1. BRACING REACTIONS ARE TO BE CONSIDERED WITH THE MAIN FRAME REACTIONS.
2. STANDARD AIC ANCHOR BOLTS CONSIST OF 2 NUTS + ONE WASHER.
3. BOTTOM OF BASE PLATES ARE AT THE SAME ELEVATION (UNLESS NOTED OTHERWISE).
4. 50mm GROUT IS REQUIRED AT ALL BASE OF COLUMN.
5. 40 x 40 CONCRETE NOTCH IS FOR SINGLE SKIN PANEL ONLY. MAY VARY DEPENDING ON THE ACTUAL WALL PANEL CONDITION.
6. ALL DIMENSIONS ARE IN MILLIMETERS.



ANCHOR BOLT DETAIL



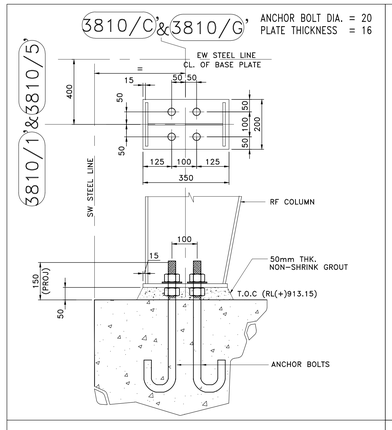
BASE PLATE TO JAMB



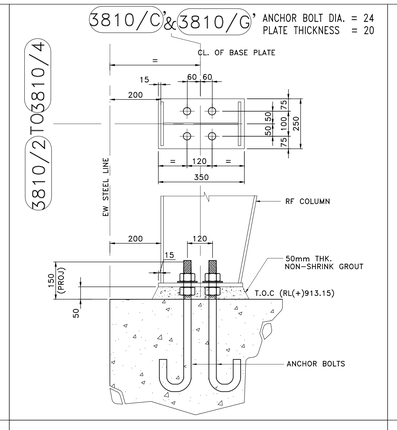
DETAIL-G

ANCHOR BOLT PLAN AT +913.05/+913.15 (F.F.L)/+912.65 (F.G.L)

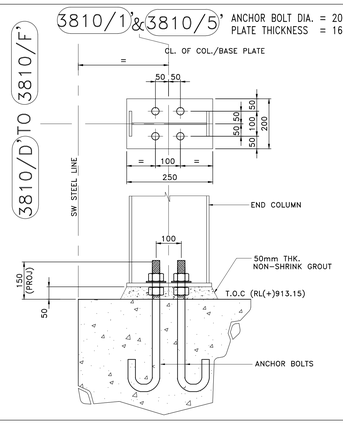
F01 - DENOTES FRAMED OPENING FOR RUD (4000WX4125H)- 2 NO'S.
F02 - DENOTES FRAMED OPENING FOR RUD (3180WX3700H)- 1 NO'S.
XB - DENOTES DIAGONAL BRACING.



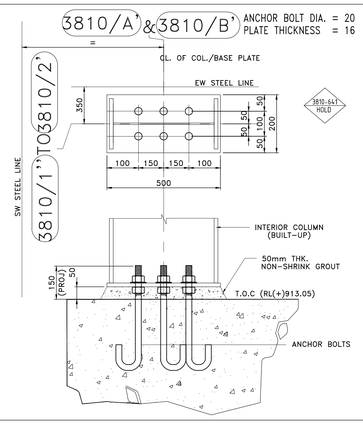
DETAIL-A



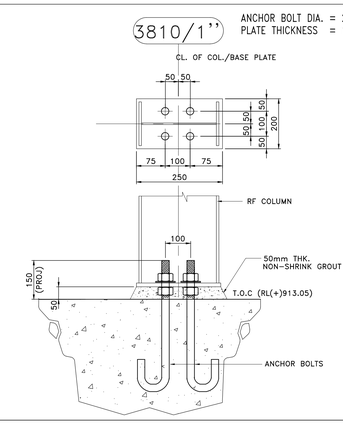
DETAIL-B



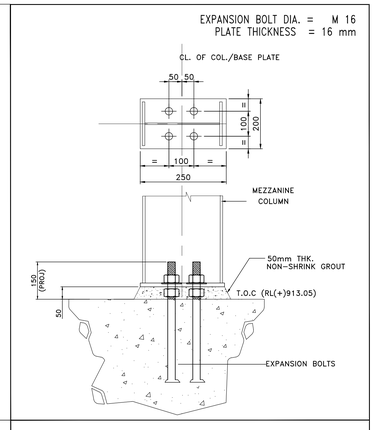
DETAIL-C



DETAIL-D



DETAIL-E



DETAIL-F

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3810-EG-ME-GAD-1011-A01	LOAD PLAN REAGENTS - CAUSTIC AND CYANIDE PLAN VIEW								
MD-216-3810-EG-ST-GAD-1001-A01	STEEL STRUCTURE GLEE DRAWING CAUSTIC SODA AND CYANIDE PLAN VIEW, 3D VIEW AND SECTIONS								
MD-216-3810-EG-ST-GAD-1001-A01	ELE inputs	A01	30/06/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

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Arabian International Company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel. : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER						REVISION
MD-P20131-B04-A001						01
DATE	23/06/2020	DATE	23/06/2020	DATE	23/06/2020	
DRAWN	7015	CHECKED	7013/7028	APPROVED	7001	

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_450_S_G_26067_001		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707_GAU01_AA24_MTL01_38103	OU100676475	0

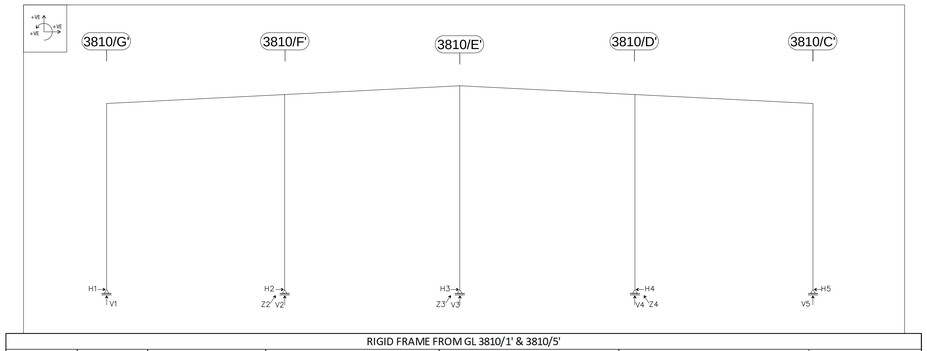
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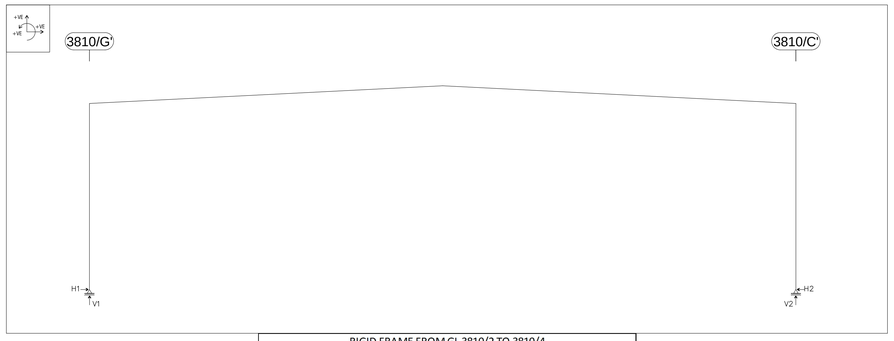
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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	27/06/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	CAUSTIC & CYAN. STORG. & MIXING STN.
APPROVED	PRI	30/06/2020	ANCHOR BOLT PLAN AND DETAILS
EMPLOYER			
SCALE 1:100	A0		DRAWING NUMBER: MD-216-3810-EG-ST-GAD-2343 REV: A01

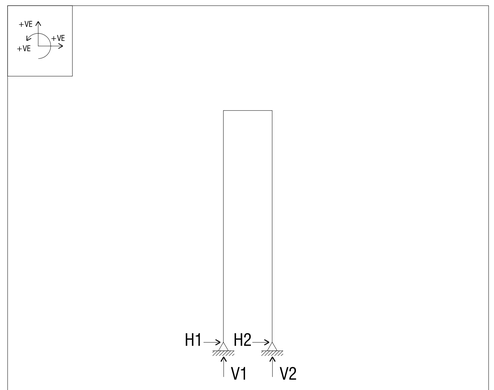
DRAWING NUMBER MD-216-3810-EG-ST-GAD-2344 REV A01

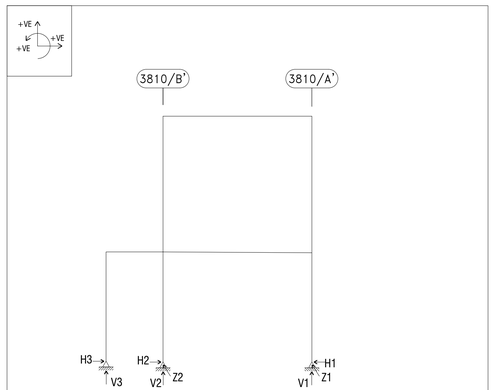


RIGID FRAME FROM GL 3810/1' & 3810/5'														
CATEGORY	LOAD CASE	LEFT COL. @GL G'		LEFT COL. @GL F'			COL. @GL E'			COL. @GL D'			RIGHT COL. @GL C'	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)	Z3 (KN)	H4 (KN)	V4 (KN)	Z4 (KN)	H5 (KN)	V5 (KN)
Dead	Dead	0.40	-4.60	0.00	-5.90	0.00	0.00	-5.90	0.00	0.00	-5.90	0.00	0.40	-4.60
Collateral	Collateral	0.10	-5.30	0.00	-0.20	0.00	0.00	-1.40	0.00	-0.20	-0.10	0.00	0.10	-0.10
Roof Live	Live	1.00	-7.20	0.00	-13.20	0.00	0.00	-12.80	0.00	0.00	-13.20	0.00	-1.00	-7.20
Wind	Wind_Left1	-15.40	27.90	0.00	31.30	22.30	0.00	28.30	23.40	0.00	28.60	22.30	-12.60	5.90
	Wind_Right1	12.60	5.90	0.00	28.60	-24.60	0.00	28.30	-25.80	0.00	31.30	-24.60	15.30	27.90
	Wind_Left2	-18.50	20.30	0.00	21.50	-24.60	0.00	18.40	23.40	0.00	18.80	22.30	-9.50	-1.70
	Wind_Right2	9.50	-1.70	0.00	18.80	22.30	0.00	18.40	-25.80	0.00	21.50	-24.60	18.50	20.20
	Wind_Long1	6.20	15.90	0.00	26.80	22.30	0.00	19.70	23.40	0.00	12.60	22.30	-4.50	13.60
	Wind_Long2	4.50	13.60	0.00	12.60	-24.60	0.00	19.70	-25.80	0.00	26.80	-24.60	-6.20	15.90
Seismic	Seismic_L	-0.30	0.40	0.00	-0.50	0.00	0.00	0.40	0.00	0.00	0.50	0.00	-0.40	-0.40
	Seismic_R	0.30	-0.40	0.00	0.50	0.00	0.00	0.00	0.00	0.00	-0.50	0.00	0.40	0.40
Temperature	25Degree	1.90	2.30	0.00	2.70	0.00	0.00	0.90	0.00	0.00	2.70	0.00	1.90	-2.30
	25Degree/-VE	-1.90	-2.30	0.00	-2.70	0.00	0.00	-0.90	0.00	0.00	-2.70	0.00	1.90	-2.30



RIGID FRAME FROM GL 3810/2 TO 3810/4					
CATEGORY	LOAD CASE	LEFT COL. @GL G'		MEZZ COL. @GL C'	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)
Dead	Dead	10.00	-16.70	-10.00	-16.70
Collateral	Collateral	-0.10	-2.10	0.10	-0.10
Roof Live	Live	34.40	-46.70	-34.40	-46.70
Wind	Wind_Left1	-69.70	89.90	37.00	64.80
	Wind_Right1	-37.00	64.80	69.70	89.90
	Wind_Left2	-50.40	50.70	17.60	25.60
	Wind_Right2	-17.60	25.60	50.40	50.70
	Wind_Long1	-38.70	118.50	41.70	102.20
	Wind_Long2	-41.70	102.20	38.70	118.50
Seismic	Seismic_L	0.50	0.20	-0.40	-0.20
	Seismic_R	-0.50	-0.20	0.40	0.20
Temperature	25Degree	-0.40	0.00	-0.40	0.00
	25Degree/-VE	0.40	0.00	0.40	0.00





RIGID FRAME @ GL 3810/2"					
CATEGORY	LOAD CASE	LEFT COLUMN		RIGHT COLUMN	
		H1 (KN)	V1 (KN)	H2 (KN)	V2 (KN)
Dead	Dead	0.00	-3.57	-0.01	-3.44
LIVE LOAD	LL	0.00	-4.17	-0.02	-4.02
CRANE	CR1	0.00	-0.70	-0.14	0.70
	CR2	0.00	-0.47	-0.09	0.47
	CR3	0.00	-1.06	-0.21	1.06
	CR4	0.00	-0.90	-0.18	0.90
Wind	Wind Left1	0.00	-9.56	-1.90	9.56
	Wind Left2	0.00	-9.60	-1.91	9.60
	Wind Right1	3.44	7.87	-1.88	-7.87
	Wind Right2	3.44	7.87	-1.88	-7.87
	Wind+Long1	0.00	-0.12	-0.02	0.12
	Wind+Long2	0.05	0.10	-0.03	-0.10
	Wind-Long1	0.00	-0.12	-0.02	0.12
	Wind-Long2	0.05	0.10	-0.03	-0.10
Seismic	S+X	0.00	-1.41	-0.28	1.41
	S-X	0.50	1.15	-0.27	-1.15
	S-Z	0.00	0.01	0.00	-0.01
	S+Z	0.00	-0.01	0.00	0.01
Temperature	25Degree	0.00	-0.11	-0.02	0.11
	25Degree/-VE	0.00	-0.06	-0.01	0.06

RIGID FRAME ALONG GL 3810/1" & 3810/ 2'									
CATEGORY	LOAD CASE	LEFT COL. @GL A'			MEZZ COL. @GL B'			STAIR COL.	
		H1 (KN)	V1 (KN)	Z1 (KN)	H2 (KN)	V2 (KN)	Z2 (KN)	H3 (KN)	V3 (KN)
Dead	Dead	0.13	-18.46	0.00	-0.04	-26.56	0.00	-0.01	-7.71
Roof Live	Live	-0.09	-5.05	0.00	0.02	-4.69	0.00	0.07	-0.21
LIVE LOAD	LL	0.71	-15.33	0.00	-0.61	-25.73	-0.01	-0.02	-8.31
CRANE	CR1	-0.90	-14.52	0.00	-2.15	-10.29	0.00	0.00	-10.39
	CR2	-1.30	-32.91	0.00	-2.22	-13.90	2.58	0.00	-12.20
	CR3	-0.39	7.30	1.21	-2.03	-5.85	0.00	0.00	-8.12
	CR4	-0.63	-3.44	0.00	-2.07	-7.97	0.00	0.00	-9.16
Wind	Wind Left1	-6.71	53.98	7.58	-25.61	43.72	0.98	0.00	-93.48
	Wind Left2	-6.36	50.20	6.39	-25.75	39.86	0.42	0.00	-92.34
	Wind Right1	2.56	-61.22	0.99	-14.96	-23.90	8.32	47.15	92.26
	Wind Right2	2.97	-64.26	0.42	-15.16	-26.78	7.63	46.81	91.61
	Wind+Long1	-0.98	81.06	35.84	0.08	77.44	32.64	1.19	1.12
	Wind+Long2	-0.60	77.88	35.84	0.03	73.85	32.61	0.85	1.36
	Wind-Long1	-0.98	81.06	35.84	0.08	77.44	32.64	1.19	1.12
	Wind-Long2	-0.60	77.88	35.84	0.03	73.85	32.61	0.85	1.36
Seismic	S+X	-1.88	11.02	1.29	-6.32	10.53	0.00	0.00	-21.99
	S-X	0.82	-14.39	0.00	-3.44	-7.88	1.74	11.50	22.50
	S-Z	0.02	-13.42	0.00	-0.06	-12.34	0.00	0.00	-0.40
	S+Z	-0.01	13.34	6.97	-0.04	12.63	6.36	0.08	0.16
Temperature	25Degree	0.43	-0.50	-0.01	-0.35	-1.79	-0.02	0.00	-1.22
	25Degree/-VE	-0.66	-0.37	8.21	-0.74	1.82	10.11	1.58	2.99

LOADS LEGEND	
Wind_Left1	Denotes wind load from left to right with internal pressure
Wind_Right1	Denotes wind load from right to left with internal pressure
Wind_Left2	Denotes wind load from left to right with internal suction
Wind_Right2	Denotes wind load from right to left with internal suction
Wind_Long1	Denotes wind load parallel to ridge from end wall with internal pressure
Wind_Long2	Denotes wind load parallel to ridge from end wall with internal suction
Seismic_L	Denotes seismic load from left
Seismic_R	Denotes seismic load from right

IMPORTANT NOTES :

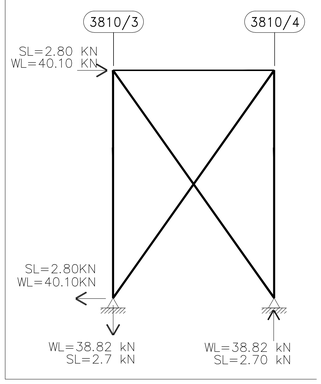
(1) BEARING CAPACITY OF CONCRETE UNDER BASE PLATE SHOULD NOT BE LESS THAN 1.05 KN/CM2

(2) GIVEN REACTIONS ARE FOR UNFACTORED LOADS.

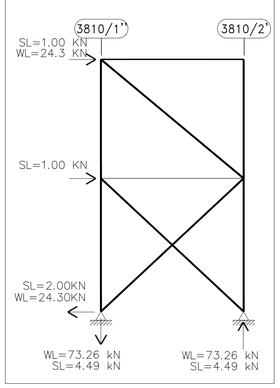
(3) BRACING REACTIONS TO BE ADDED WITH MAIN FRAME COLUMN REACTIONS FOR THE DESIGN OF FOUNDATION.

(4) LOADS IN THE SAME CATEGORY , NOT TO BE COMBINED IN THE SAME LOAD COMBINATION .

SIDEWALL (XB) BRACING REACTION AT GL 3810/G' & 3810/C'



SIDEWALL (PB) BRACING REACTION AT GL 3810/A' & 3810/B'



Building reactions are based on the following building data:

ROOF LOADS			
DEAD LOAD (kN/m2)		=	0.10
ROOF LIVE LOAD (kN/m2)		=	1.00 REDUCIBLE AS PER SBC
BASIC LOADS			
WIND LOAD	Wind Speed (3 sec)	=	155 (Km/Hr)
	Wind Code	=	ASCE (7-16)
	EXPOSURE	=	C
	CLASSIFICATION	=	ENCLOSED
SEIS LOAD IBC 2006	Ss	=	10%g
	S1	=	2%g
	SITE CLASS	=	B
TEMPERATURE LOAD	TEMPERATURE LOAD	=	± 25 C
PLATFORM LOADS			
LIVE LOAD (kN/m2)		=	3.00
DEAD LOAD (kN/m2)		=	0.50

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3810-EG-ME-GAD-1011-A01	LOAD PLAN REAGENTS - CAUSTIC AND CYANIDE PLAN VIEW								
MD-216-3810-EG-ST-GAD-1001-A01	STEEL STRUCTURE GUIDE DRAWING CAUSTIC SODA AND CYANIDE PLAN VIEW, 3D VIEW AND SECTIONS								
MD-216-3810-EG-ST-GAD-1001-A01	ELE inputs	A01	30/06/2020	ISSUED FOR REFERENCE	--	SYC	--	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

aic — arabian international company
P.O.Box : 41017 Jeddah 21521,K.S.A
Tel : +966 2 591 0999
Fax : +966 2 591 0099
E-mail : aic@aicsteel.com
Site : www.aicsteel.com

DRAWING NUMBER MD-P20131-B04-A002 — REVISION 01
DATE 23/06/2020 | DRAWN 7015 | CHECKED 7013/7028 | APPROVED 7001

LARSEN & TOUBRO DRAWING NUMBER 019045_450_S_G_26067_002 — REVISION 0
OUTOTEC DRAWING NUMBER 900707_GAU01_AA24_MTL01_38104 — DOCUMENT ID OU1006764.76 — REVISION 0

LARSEN & TOUBRO

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DRAWN	--	--
CHECKED	SYC	25/06/2020
ENGINEER	--	--
APPROVED	PRI	30/06/2020
EMPLOYER		
SCALE	N.T.S	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
CAUSTIC & CYAN. STORG. & MIXING STN.
COLUMN REACTIONS

DRAWING NUMBER MD-216-3810-EG-ST-GAD-2344 — REV A01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/07/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
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410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397;
413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407;
422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415;
430 309 311 417 416; 431 315 316 419 418; 432 316 10 409 419;
433 314 420 421 311; 434 420 422 423 421; 435 329 424 420 314;
436 424 425 422 420; 437 340 426 424 329; 438 426 427 425 424;
439 351 428 426 340; 440 428 429 427 426; 441 362 430 428 351;
442 430 431 429 428; 443 381 432 433 382; 444 432 434 435 433;
445 395 436 432 381; 446 436 437 434 432; 447 406 438 436 395;
448 438 439 437 436; 449 417 440 438 406; 450 440 441 439 438;
451 311 421 440 417; 452 421 423 441 440; 453 442 443 444 445;
454 443 319 315 444; 455 446 447 443 442; 456 447 330 319 443;
457 448 449 447 446; 458 449 341 330 447; 459 450 451 449 448;
460 451 352 341 449; 461 452 453 451 450; 462 453 363 352 451;
463 454 455 456 457; 464 455 383 386 456; 465 458 459 455 454;
466 459 396 383 455; 467 460 461 459 458; 468 461 407 396 459;
469 462 463 461 460; 470 463 418 407 461; 471 445 444 463 462;
472 444 315 418 463; 473 464 465 355 354; 474 465 466 356 355;
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472 444 315 418 463; 473 464 465 355 354; 474 465 466 356 355;
475 466 467 357 356; 476 467 468 358 357; 477 468 469 359 358;
478 469 470 360 359; 479 470 471 361 360; 480 471 472 362 361;
481 473 474 364 363; 482 474 464 354 364; 483 475 476 465 464;
484 476 477 466 465; 485 477 478 467 466; 486 478 479 468 467;
487 479 480 469 468; 488 480 481 470 469; 489 481 482 471 470;
490 482 483 472 471; 491 484 485 474 473; 492 485 475 464 474;
493 472 486 430 362; 494 486 487 431 430; 495 483 488 486 472;
496 488 489 487 486; 497 490 491 453 452; 498 491 473 363 453;
499 492 493 491 490; 500 493 484 473 491; 501 494 495 496 497;
502 495 498 499 496; 503 498 500 501 499; 504 500 502 503 501;
505 502 504 505 503; 506 504 506 507 505; 507 506 508 509 507;
508 508 510 511 509; 509 512 513 514 515; 510 513 494 497 514;
511 368 367 495 494; 512 367 370 498 495; 513 370 372 500 498;
514 372 374 502 500; 515 374 376 504 502; 516 376 378 506 504;
517 378 380 508 506; 518 380 382 510 508; 519 386 385 513 512;
520 385 368 494 513; 521 510 516 517 511; 522 516 518 519 517;
523 382 433 516 510; 524 433 435 518 516; 525 520 521 522 523;
526 521 512 515 522; 527 457 456 521 520; 528 456 386 512 521;
529 524 442 445 525; 530 526 446 442 524; 531 527 448 446 526;
532 528 450 448 527; 533 529 452 450 528; 534 530 454 457 531;
535 532 458 454 530; 536 533 460 458 532; 537 534 462 460 533;
538 525 445 462 534; 539 535 490 452 529; 540 536 492 490 535;
541 537 520 523 538; 542 531 457 520 537; 543 422 539 540 423;
544 425 541 539 422; 545 427 542 541 425; 546 429 543 542 427;
547 431 544 543 429; 548 434 545 546 435; 549 437 547 545 434;
550 439 548 547 437; 551 441 549 548 439; 552 423 540 549 441;
553 487 550 544 431; 554 489 551 550 487; 555 518 552 553 519;
556 435 546 552 518; 557 497 496 554 555; 558 496 499 556 554;
559 499 501 557 556; 560 501 503 558 557; 561 503 505 559 558;
562 505 507 560 559; 563 507 509 561 560; 564 509 511 562 561;
565 515 514 563 564; 566 514 497 555 563; 567 511 517 565 562;
568 517 519 566 565; 569 523 522 567 568; 570 522 515 564 567;
571 538 523 568 569; 572 519 553 570 566; 573 571 572 476 475;
574 572 573 477 476; 575 573 574 478 477; 576 574 575 479 478;
577 575 576 480 479; 578 576 577 481 480; 579 577 578 482 481;
580 578 579 483 482; 581 580 581 485 484; 582 581 571 475 485;
583 579 582 488 483; 584 582 583 489 488; 585 584 585 493 492;
586 585 580 484 493; 587 586 584 492 536; 588 583 587 551 489;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 588
_RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 588
_RAFT_SHEAR 337 338 341 342 349 TO 352 359 TO 402 409 TO 412 419 TO 422 429 -
430 TO 588
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 588 THICKNESS 0.6
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 587 -
588 PLATE MAT DIRECT Y SUBGRADE 90000 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -18
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -20
JOINT LOAD
128 FX 10 FY -16.7
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE COLATERAL LOAD ( CL )
JOINT LOAD
```



```
JOINT LOAD
128 FX 0.1 FY -2.1
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 34.4 FY -46.7
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX -69.7 FY 128.72 FZ 40.1
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX -37 FY 103.62 FZ 40.1
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX -50.4 FY 89.52 FZ 40.1
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX -17.6 FY 64.42 FZ 40.1
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -38.7 FY 157.32 FZ 40.1
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -41.7 FY 141.02 FZ 40.1
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 0.5 FY 2.9 FZ 2.8
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -0.5 FY 2.5 FZ 2.8
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FX -0.4
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX 0.4
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL + CL
1 1.0 2 1.0
LOAD COMB 202 DL+CL+L+TL1
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 203 DL+CL+L+TL2
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 204 DL+CL+0.75L+0.75TL1
1 1.0 2 1.0 3 0.75 12 0.75
LOAD COMB 205 DL+CL+0.75L+0.75TL2
1 1.0 2 1.0 3 0.75 13 0.75
LOAD COMB 206 DL+CL+WL1
1 1.0 2 1.0 4 1.0
LOAD COMB 207 DL+CL+WL2
1 1.0 2 1.0 5 1.0
LOAD COMB 208 DL+CL+WL3
1 1.0 2 1.0 6 1.0
LOAD COMB 209 DL+CL+WL4
1 1.0 2 1.0 7 1.0
LOAD COMB 210 DL+CL+WL5
1 1.0 2 1.0 8 1.0
LOAD COMB 211 DL+CL+WL6
1 1.0 2 1.0 9 1.0
LOAD COMB 212 DL+CL+0.75WL1+0.75L
1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
LOAD COMB 213 DL+CL+0.75WL2+0.75L
1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
LOAD COMB 214 DL+CL+0.75WL3+0.75L
1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
LOAD COMB 215 DL+CL+0.75WL4+0.75L
1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
LOAD COMB 216 DL+CL+0.75WL5+0.75L
```

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LOAD COMB 216 DL+CL+0.75WL5+0.75L
1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
LOAD COMB 217 DL+CL+0.75WL67+0.75L
1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
LOAD COMB 218 0.6DL+CL+WL1
1 0.6 2 1.0 4 1.0
LOAD COMB 219 0.6DL+CL+WL2
1 0.6 2 1.0 5 1.0
LOAD COMB 220 0.6DL+CL+WL3
1 0.6 2 1.0 6 1.0
LOAD COMB 221 0.6DL+CL+WL4
1 0.6 2 1.0 7 1.0
LOAD COMB 222 0.6DL+CL+WL5
1 0.6 2 1.0 8 1.0
LOAD COMB 223 0.6DL+CL+WL6
1 0.6 2 1.0 9 1.0
LOAD COMB 224 0.6DL+CL+0.7SL1
1 0.6 2 1.0 10 0.7
LOAD COMB 225 0.6DL+CL+0.7SL2
1 0.6 2 1.0 11 0.7
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL+1.4CL
1 1.4 2 1.4
LOAD COMB 102 1.2DL+1.2CL+1.2TL1+1.6L
1 1.2 2 1.2 3 1.6 12 1.2
LOAD COMB 103 1.2DL+1.2CL+1.2TL2+1.6L
1 1.2 2 1.2 3 1.6 13 1.2
LOAD COMB 104 1.2DL+1.2CL+0.5L
1 1.2 2 1.2 3 0.5
LOAD COMB 105 1.2DL+1.2CL+1.6WL1+0.5L
1 1.2 2 1.2 4 1.6 3 0.5
LOAD COMB 106 1.2DL+1.2CL+1.6WL2+0.5L
1 1.2 2 1.2 5 1.6 3 0.5
LOAD COMB 107 1.2DL+1.2CL+1.6WL3+0.5L
1 1.2 2 1.2 6 1.6 3 0.5
LOAD COMB 108 1.2DL+1.2CL+1.6WL4+0.5L
1 1.2 2 1.2 7 1.6 3 0.5
LOAD COMB 109 1.2DL+1.2CL+1.6WL5+0.5L
1 1.2 2 1.2 8 1.6 3 0.5
LOAD COMB 110 1.2DL+1.2CL+1.6WL6+0.5L
1 1.2 2 1.2 9 1.6 3 0.5
LOAD COMB 111 1.2DL+1.2CL+1.0SL1+0.5L
1 1.2 2 1.2 3 0.5 10 1.0
LOAD COMB 112 1.2DL+1.2CL+1.0SL2+0.5L
1 1.2 2 1.2 3 0.5 11 1.0
LOAD COMB 113 0.9DL+0.9CL+1.6WL1
1 0.9 2 0.9 4 1.6
LOAD COMB 114 0.9DL+0.9CL+1.6WL2
1 0.9 2 0.9 5 1.6
LOAD COMB 115 0.9DL+0.9CL+1.6WL3
1 0.9 2 0.9 6 1.6
LOAD COMB 116 0.9DL+0.9CL+1.6WL4
1 0.9 2 0.9 7 1.6
LOAD COMB 117 0.9DL+0.9CL+1.6WL5
1 0.9 2 0.9 8 1.6
LOAD COMB 118 0.9DL+0.9CL+1.6WL6
1 0.9 2 0.9 9 1.6
LOAD COMB 119 0.9DL+0.9CL+1.0SL1
1 0.9 2 0.9 10 1.0
LOAD COMB 120 0.9DL+0.9CL+1.0SL2
1 0.9 2 0.9 11 1.0
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 120
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 338 341 TO 432
MAXMAIN 20 ALL
MINMAIN 20 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/07/2020
JOB NAME FDN. MKD. F1_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 912.2 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 10 FY -16.7
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE COLATERAL LOAD ( CL )
JOINT LOAD
128 FX 0.1 FY -2.1
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FX 34.4 FY -46.7
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FX -69.7 FY 128.72 FZ 40.1
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FX -37 FY 103.62 FZ 40.1
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FX -50.4 FY 89.52 FZ 40.1
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FX -17.6 FY 64.42 FZ 40.1
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FX -38.7 FY 157.32 FZ 40.1
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FX -41.7 FY 141.02 FZ 40.1
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FX 0.5 FY 2.9 FZ 2.8
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FX -0.5 FY 2.5 FZ 2.8
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
```



```
JOINT LOAD
128 FX -0.4
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FX 0.4
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 DL + CL
1 1.0 2 1.0
LOAD COMB 202 DL+CL+L+TL1
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 203 DL+CL+L+TL2
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 204 DL+CL+0.75L+0.75TL1
1 1.0 2 1.0 3 0.75 12 0.75
LOAD COMB 205 DL+CL+0.75L+0.75TL2
1 1.0 2 1.0 3 0.75 13 0.75
LOAD COMB 206 DL+CL+WL1
1 1.0 2 1.0 4 1.0
LOAD COMB 207 DL+CL+WL2
1 1.0 2 1.0 5 1.0
LOAD COMB 208 DL+CL+WL3
1 1.0 2 1.0 6 1.0
LOAD COMB 209 DL+CL+WL4
1 1.0 2 1.0 7 1.0
LOAD COMB 210 DL+CL+WL5
1 1.0 2 1.0 8 1.0
LOAD COMB 211 DL+CL+WL6
1 1.0 2 1.0 9 1.0
LOAD COMB 212 DL+CL+0.75WL1+0.75L
1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
LOAD COMB 213 DL+CL+0.75WL2+0.75L
1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
LOAD COMB 214 DL+CL+0.75WL3+0.75L
1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
LOAD COMB 215 DL+CL+0.75WL4+0.75L
1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
LOAD COMB 216 DL+CL+0.75WL5+0.75L
1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
LOAD COMB 217 DL+CL+0.75WL67+0.75L
1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
LOAD COMB 218 0.6DL+CL+WL1
1 0.6 2 1.0 4 1.0
LOAD COMB 219 0.6DL+CL+WL2
1 0.6 2 1.0 5 1.0
LOAD COMB 220 0.6DL+CL+WL3
1 0.6 2 1.0 6 1.0
LOAD COMB 221 0.6DL+CL+WL4
1 0.6 2 1.0 7 1.0
LOAD COMB 222 0.6DL+CL+WL5
1 0.6 2 1.0 8 1.0
LOAD COMB 223 0.6DL+CL+WL6
1 0.6 2 1.0 9 1.0
LOAD COMB 224 0.6DL+CL+0.7SL1
1 0.6 2 1.0 10 0.7
LOAD COMB 225 0.6DL+CL+0.7SL2
1 0.6 2 1.0 11 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/07/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175;
35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175;
128 0 912.2 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438;
304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438;
307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175;
311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438;
315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438;
320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438;
323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438;
326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438;
329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438;
332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438;
335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438;
338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438;
341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438;
344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438;
347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438;
350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438;
353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438;
356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438;
359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438;
362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438;
365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925;
368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925;
371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438;
374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925;
377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438;
380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925;
383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925;
386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438;
389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438;
392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438;
395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438;
398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438;
401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438;
404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438;
407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438;
410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438;
413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438;
416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438;
419 -1 909.9 2.92438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309;
341 319 320 316 315; 342 320 303 10 316; 343 321 322 302 303;
344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305;
347 325 326 307 306; 348 326 327 308 307; 349 327 328 313 308;
350 328 329 314 313; 351 330 331 320 319; 352 331 321 303 320;
353 332 333 322 321; 354 333 334 323 322; 355 334 335 324 323;
356 335 336 325 324; 357 336 337 326 325; 358 337 338 327 326;
359 338 339 328 327; 360 339 340 329 328; 361 341 342 331 330;
362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333;
365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336;
368 348 349 338 337; 369 349 350 339 338; 370 350 351 340 339;
371 352 353 342 341; 372 353 343 332 342; 373 354 355 344 343;
374 355 356 345 344; 375 356 357 346 345; 376 357 358 347 346;
377 358 359 348 347; 378 359 360 349 348; 379 360 361 350 349;
380 361 362 351 350; 381 363 364 353 352; 382 364 354 343 353;
383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370;
386 371 373 374 372; 387 373 375 376 374; 388 375 377 378 376;
389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386;
392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366;
395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373;
398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379;
401 396 397 384 383; 402 397 387 365 384; 403 398 399 388 387;
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401 396 397 384 383; 402 397 387 365 384; 403 398 399 388 387;
404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393;
410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397;
413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407;
422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415;
430 309 311 417 416; 431 315 316 419 418; 432 316 10 409 419;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 432
_RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 432
_RAFT_SHEAR 326 334 337 338 341 TO 343 348 TO 413 418 TO 423 428 TO 432
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 432 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 431 -
432 PLATE MAT DIRECT Y SUBGRADE 90000 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 432 PR GY -20
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 432 PR GY -20
JOINT LOAD
128 FY -5.9
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE COLATERAL LOAD ( CL )
JOINT LOAD
128 FY -0.2
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -13.2
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 31.3 FZ 22.3
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 28.6 FZ 24.6
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 21.5 FZ 24.6
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 18.8 FZ 22.3
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 26.8 FZ 22.3
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY 12.6 FZ 24.6
*********************************************************************************************
```

```
**********************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -0.5
**********************************************************************************************
LOAD 11 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 0.5
**********************************************************************************************
LOAD 12 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
128 FY 2.7
**********************************************************************************************
LOAD 13 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -2.7
**********************************************************************************************
* UNFACTORED LOAD COMBINATIONS
LOAD COMB 201 DL + CL
1 1.0 2 1.0
LOAD COMB 202 DL+CL+L+TL1
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 203 DL+CL+L+TL2
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 204 DL+CL+0.75L+0.75TL1
1 1.0 2 1.0 3 0.75 12 0.75
LOAD COMB 205 DL+CL+0.75L+0.75TL2
1 1.0 2 1.0 3 0.75 13 0.75
LOAD COMB 206 DL+CL+WL1
1 1.0 2 1.0 4 1.0
LOAD COMB 207 DL+CL+WL2
1 1.0 2 1.0 5 1.0
LOAD COMB 208 DL+CL+WL3
1 1.0 2 1.0 6 1.0
LOAD COMB 209 DL+CL+WL4
1 1.0 2 1.0 7 1.0
LOAD COMB 210 DL+CL+WL5
1 1.0 2 1.0 8 1.0
LOAD COMB 211 DL+CL+WL6
1 1.0 2 1.0 9 1.0
LOAD COMB 212 DL+CL+0.75WL1+0.75L
1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
LOAD COMB 213 DL+CL+0.75WL2+0.75L
1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
LOAD COMB 214 DL+CL+0.75WL3+0.75L
1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
LOAD COMB 215 DL+CL+0.75WL4+0.75L
1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
LOAD COMB 216 DL+CL+0.75WL5+0.75L
1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
LOAD COMB 217 DL+CL+0.75WL67+0.75L
1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
LOAD COMB 218 0.6DL+CL+WL1
1 0.6 2 1.0 4 1.0
LOAD COMB 219 0.6DL+CL+WL2
1 0.6 2 1.0 5 1.0
LOAD COMB 220 0.6DL+CL+WL3
1 0.6 2 1.0 6 1.0
LOAD COMB 221 0.6DL+CL+WL4
1 0.6 2 1.0 7 1.0
LOAD COMB 222 0.6DL+CL+WL5
1 0.6 2 1.0 8 1.0
LOAD COMB 223 0.6DL+CL+WL6
1 0.6 2 1.0 9 1.0
LOAD COMB 224 0.6DL+CL+0.7SL1
1 0.6 2 1.0 10 0.7
LOAD COMB 225 0.6DL+CL+0.7SL2
1 0.6 2 1.0 11 0.7
**********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL+1.4CL
1 1.4 2 1.4
LOAD COMB 102 1.2DL+1.2CL+1.2TL1+1.6L
1 1.2 2 1.2 3 1.6 12 1.2
LOAD COMB 103 1.2DL+1.2CL+1.2TL2+1.6L
1 1.2 2 1.2 3 1.6 13 1.2
LOAD COMB 104 1.2DL+1.2CL+0.5L
1 1.2 2 1.2 3 0.5
LOAD COMB 105 1.2DL+1.2CL+1.6WL1+0.5L
```



```
LOAD COMB 105 1.2DL+1.2CL+1.6WL1+0.5L
1 1.2 2 1.2 4 1.6 3 0.5
LOAD COMB 106 1.2DL+1.2CL+1.6WL2+0.5L
1 1.2 2 1.2 5 1.6 3 0.5
LOAD COMB 107 1.2DL+1.2CL+1.6WL3+0.5L
1 1.2 2 1.2 6 1.6 3 0.5
LOAD COMB 108 1.2DL+1.2CL+1.6WL4+0.5L
1 1.2 2 1.2 7 1.6 3 0.5
LOAD COMB 109 1.2DL+1.2CL+1.6WL5+0.5L
1 1.2 2 1.2 8 1.6 3 0.5
LOAD COMB 110 1.2DL+1.2CL+1.6WL6+0.5L
1 1.2 2 1.2 9 1.6 3 0.5
LOAD COMB 111 1.2DL+1.2CL+1.0SL1+0.5L
1 1.2 2 1.2 3 0.5 10 1.0
LOAD COMB 112 1.2DL+1.2CL+1.0SL2+0.5L
1 1.2 2 1.2 3 0.5 11 1.0
LOAD COMB 113 0.9DL+0.9CL+1.6WL1
1 0.9 2 0.9 4 1.6
LOAD COMB 114 0.9DL+0.9CL+1.6WL2
1 0.9 2 0.9 5 1.6
LOAD COMB 115 0.9DL+0.9CL+1.6WL3
1 0.9 2 0.9 6 1.6
LOAD COMB 116 0.9DL+0.9CL+1.6WL4
1 0.9 2 0.9 7 1.6
LOAD COMB 117 0.9DL+0.9CL+1.6WL5
1 0.9 2 0.9 8 1.6
LOAD COMB 118 0.9DL+0.9CL+1.6WL6
1 0.9 2 0.9 9 1.6
LOAD COMB 119 0.9DL+0.9CL+1.0SL1
1 0.9 2 0.9 10 1.0
LOAD COMB 120 0.9DL+0.9CL+1.0SL2
1 0.9 2 0.9 11 1.0
**********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 120
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 20 ALL
MINMAIN 20 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 04/07/2020
JOB NAME FDN. MKD. F2_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 912.2 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FY -5.9
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE COLATERAL LOAD ( CL )
JOINT LOAD
128 FY -0.2
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE LIVE LOAD ( L )
JOINT LOAD
128 FY -13.2
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD  ( WL1 )
JOINT LOAD
128 FY 31.3 FZ 22.3
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD  ( WL2 )
JOINT LOAD
128 FY 28.6 FZ 24.6
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD  ( WL3 )
JOINT LOAD
128 FY 21.5 FZ 24.6
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE WIND LOAD  ( WL4 )
JOINT LOAD
128 FY 18.8 FZ 22.3
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE WIND LOAD  ( WL5 )
JOINT LOAD
128 FY 26.8 FZ 22.3
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE WIND LOAD  ( WL6 )
JOINT LOAD
128 FY 12.6 FZ 24.6
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( SL1 )
JOINT LOAD
128 FY -0.5
*********************************************************************************************
LOAD 11 LOADTYPE None  TITLE SEISMIC LOAD  ( SL2 )
JOINT LOAD
128 FY 0.5
*********************************************************************************************
LOAD 12 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL1 )
JOINT LOAD
```



```
JOINT LOAD
128 FY 2.7
*********************************************************************************************
LOAD 13 LOADTYPE None  TITLE TEMPERATURE LOAD  ( TL2 )
JOINT LOAD
128 FY -2.7
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 DL + CL
1 1.0 2 1.0
LOAD COMB 202 DL+CL+L+TL1
1 1.0 2 1.0 3 1.0 12 1.0
LOAD COMB 203 DL+CL+L+TL2
1 1.0 2 1.0 3 1.0 13 1.0
LOAD COMB 204 DL+CL+0.75L+0.75TL1
1 1.0 2 1.0 3 0.75 12 0.75
LOAD COMB 205 DL+CL+0.75L+0.75TL2
1 1.0 2 1.0 3 0.75 13 0.75
LOAD COMB 206 DL+CL+WL1
1 1.0 2 1.0 4 1.0
LOAD COMB 207 DL+CL+WL2
1 1.0 2 1.0 5 1.0
LOAD COMB 208 DL+CL+WL3
1 1.0 2 1.0 6 1.0
LOAD COMB 209 DL+CL+WL4
1 1.0 2 1.0 7 1.0
LOAD COMB 210 DL+CL+WL5
1 1.0 2 1.0 8 1.0
LOAD COMB 211 DL+CL+WL6
1 1.0 2 1.0 9 1.0
LOAD COMB 212 DL+CL+0.75WL1+0.75L
1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
LOAD COMB 213 DL+CL+0.75WL2+0.75L
1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
LOAD COMB 214 DL+CL+0.75WL3+0.75L
1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
LOAD COMB 215 DL+CL+0.75WL4+0.75L
1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
LOAD COMB 216 DL+CL+0.75WL5+0.75L
1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
LOAD COMB 217 DL+CL+0.75WL67+0.75L
1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
LOAD COMB 218 0.6DL+CL+WL1
1 0.6 2 1.0 4 1.0
LOAD COMB 219 0.6DL+CL+WL2
1 0.6 2 1.0 5 1.0
LOAD COMB 220 0.6DL+CL+WL3
1 0.6 2 1.0 6 1.0
LOAD COMB 221 0.6DL+CL+WL4
1 0.6 2 1.0 7 1.0
LOAD COMB 222 0.6DL+CL+WL5
1 0.6 2 1.0 8 1.0
LOAD COMB 223 0.6DL+CL+WL6
1 0.6 2 1.0 9 1.0
LOAD COMB 224 0.6DL+CL+0.7SL1
1 0.6 2 1.0 10 0.7
LOAD COMB 225 0.6DL+CL+0.7SL2
1 0.6 2 1.0 11 0.7
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 4 STAAD OUTPUT – FOUNDATION F1

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    JUL 20, 2020                 *
*          Time=    18:45:16                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/07/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 912.2 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
   20. 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438
   21. 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438
   22. 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438
   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
   28. 344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438
   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
   30. 350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438
   31. 353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438
   32. 356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438
   33. 359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438
   34. 362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438
   35. 365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925
   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
   37. 371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438
   38. 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925
```



```
   39. 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438
   40. 380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925
   41. 383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925
   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
   43. 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438
   44. 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438
   45. 395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438
   46. 398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438
   47. 401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438
   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
   49. 407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438
   50. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   51. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   52. 416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438
   53. 419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   54. 422 1.75 909.9 3.42438; 423 1.75 909.9 3.175; 424 1.5 909.9 3.67438
   55. 425 1.75 909.9 3.67438; 426 1.5 909.9 3.92438; 427 1.75 909.9 3.92438
   56. 428 1.5 909.9 4.17438; 429 1.75 909.9 4.17438; 430 1.5 909.9 4.42438
   57. 431 1.75 909.9 4.42438; 432 1.5 909.9 2.17438; 433 1.5 909.9 1.925
   58. 434 1.75 909.9 2.17438; 435 1.75 909.9 1.925; 436 1.5 909.9 2.42438
   59. 437 1.75 909.9 2.42438; 438 1.5 909.9 2.67438; 439 1.75 909.9 2.67438
   60. 440 1.5 909.9 2.92438; 441 1.75 909.9 2.92438; 442 -1.75 909.9 3.42438
   61. 443 -1.5 909.9 3.42438; 444 -1.5 909.9 3.175; 445 -1.75 909.9 3.175
   62. 446 -1.75 909.9 3.67438; 447 -1.5 909.9 3.67438; 448 -1.75 909.9 3.92438
   63. 449 -1.5 909.9 3.92438; 450 -1.75 909.9 4.17438; 451 -1.5 909.9 4.17438
   64. 452 -1.75 909.9 4.42438; 453 -1.5 909.9 4.42438; 454 -1.75 909.9 2.17438
   65. 455 -1.5 909.9 2.17438; 456 -1.5 909.9 1.925; 457 -1.75 909.9 1.925
   66. 458 -1.75 909.9 2.42438; 459 -1.5 909.9 2.42438; 460 -1.75 909.9 2.67438
   67. 461 -1.5 909.9 2.67438; 462 -1.75 909.9 2.92438; 463 -1.5 909.9 2.92438
   68. 464 -0.75 909.9 4.67438; 465 -0.5 909.9 4.67438; 466 -0.25 909.9 4.67438
   69. 467 0 909.9 4.67438; 468 0.25 909.9 4.67438; 469 0.5 909.9 4.67438
   70. 470 0.75 909.9 4.67438; 471 1 909.9 4.67438; 472 1.25 909.9 4.67438
   71. 473 -1.25 909.9 4.67438; 474 -1 909.9 4.67438; 475 -0.75 909.9 4.92438
   72. 476 -0.5 909.9 4.92438; 477 -0.25 909.9 4.92438; 478 0 909.9 4.92438
   73. 479 0.25 909.9 4.92438; 480 0.5 909.9 4.92438; 481 0.75 909.9 4.92438
   74. 482 1 909.9 4.92438; 483 1.25 909.9 4.92438; 484 -1.25 909.9 4.92438
   75. 485 -1 909.9 4.92438; 486 1.5 909.9 4.67438; 487 1.75 909.9 4.67438
   76. 488 1.5 909.9 4.92438; 489 1.75 909.9 4.92438; 490 -1.75 909.9 4.67438
   77. 491 -1.5 909.9 4.67438; 492 -1.75 909.9 4.92438; 493 -1.5 909.9 4.92438
   78. 494 -0.75 909.9 1.67438; 495 -0.5 909.9 1.67438; 496 -0.5 909.9 1.425
   79. 497 -0.75 909.9 1.425; 498 -0.25 909.9 1.67438; 499 -0.25 909.9 1.425
   80. 500 0 909.9 1.67438; 501 0 909.9 1.425; 502 0.25 909.9 1.67438
   81. 503 0.25 909.9 1.425; 504 0.5 909.9 1.67438; 505 0.5 909.9 1.425
   82. 506 0.75 909.9 1.67438; 507 0.75 909.9 1.425; 508 1 909.9 1.67438
   83. 509 1 909.9 1.425; 510 1.25 909.9 1.67438; 511 1.25 909.9 1.425
   84. 512 -1.25 909.9 1.67438; 513 -1 909.9 1.67438; 514 -1 909.9 1.425
   85. 515 -1.25 909.9 1.425; 516 1.5 909.9 1.67438; 517 1.5 909.9 1.425
   86. 518 1.75 909.9 1.67438; 519 1.75 909.9 1.425; 520 -1.75 909.9 1.67438
   87. 521 -1.5 909.9 1.67438; 522 -1.5 909.9 1.425; 523 -1.75 909.9 1.425
   88. 524 -1.875 909.9 3.42438; 525 -1.875 909.9 3.175; 526 -1.875 909.9 3.67438
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196. START GROUP DEFINITION
197. ELEMENT
198. _RAFT 326 TO 328 332 TO 334 337 338 341 TO 588
199. _RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 588
200. _RAFT_SHEAR 337 338 341 342 349 TO 352 359 TO 402 409 TO 412 419 TO 422 429 -
201. 430 TO 588
202. MEMBER
203. END GROUP DEFINITION
204. ELEMENT PROPERTY
205. 326 TO 328 332 TO 334 337 338 341 TO 588 THICKNESS 0.6
206. DEFINE MATERIAL START
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207. ISOTROPIC CONCRETE
208. E 2.17185E+007
209. POISSON 0.17
210. DENSITY 25
211. ALPHA 1E-005
212. DAMP 0.05
213. TYPE CONCRETE
214. STRENGTH FCU 27579
215. END DEFINE MATERIAL
216. MEMBER PROPERTY AMERICAN
217. 185 PRIS YD 0.5 ZD 0.5
218. CONSTANTS
219. MATERIAL CONCRETE ALL
220. SUPPORTS
221. 326 TO 328 332 TO 334 337 338 341 TO 587 -
222. 588 PLATE MAT DIRECT Y SUBGRADE 90000 PRINT
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STAAD SPACE -- PAGE NO. 7

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STAAD SPACE -- PAGE NO. 8

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224. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
225. SELFWEIGHT Y -1
226. ELEMENT LOAD
227. * LOAD DUE TO BACKFILL
228. 326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -18
229. * LOAD DUE TO SURCHARGE
230. 326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -20
231. JOINT LOAD
232. 128 FX 10 FY -16.7
233. ********************************************************************************
234. LOAD 2 LOADTYPE NONE  TITLE COLATERAL LOAD ( CL )
235. JOINT LOAD
236. 128 FX 0.1 FY -2.1
237. ********************************************************************************
238. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
239. JOINT LOAD
240. 128 FX 34.4 FY -46.7
241. ********************************************************************************
242. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
243. JOINT LOAD
244. 128 FX -69.7 FY 128.72 FZ 40.1
245. ********************************************************************************
246. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
247. JOINT LOAD
248. 128 FX -37 FY 103.62 FZ 40.1
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249. ********************************************************************************
250. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
251. JOINT LOAD
252. 128 FX -50.4 FY 89.52 FZ 40.1
253. ********************************************************************************
254. LOAD 7 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
255. JOINT LOAD
256. 128 FX -17.6 FY 64.42 FZ 40.1
257. ********************************************************************************
258. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
259. JOINT LOAD
260. 128 FX -38.7 FY 157.32 FZ 40.1
261. ********************************************************************************
262. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
263. JOINT LOAD
264. 128 FX -41.7 FY 141.02 FZ 40.1
265. ********************************************************************************
266. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
267. JOINT LOAD
268. 128 FX 0.5 FY 2.9 FZ 2.8
269. ********************************************************************************
270. LOAD 11 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
271. JOINT LOAD
272. 128 FX -0.5 FY 2.5 FZ 2.8
273. ********************************************************************************
274. LOAD 12 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
275. JOINT LOAD
276. 128 FX -0.4
277. ********************************************************************************
278. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
279. JOINT LOAD
280. 128 FX 0.4
281. ********************************************************************************
282. * UNFACTORED LOAD COMBINATIONS
283. LOAD COMB 201 DL + CL
284. 1 1.0 2 1.0
285. LOAD COMB 202 DL+CL+L+TL1
286. 1 1.0 2 1.0 3 1.0 12 1.0
287. LOAD COMB 203 DL+CL+L+TL2
288. 1 1.0 2 1.0 3 1.0 13 1.0
289. LOAD COMB 204 DL+CL+0.75L+0.75TL1
290. 1 1.0 2 1.0 3 0.75 12 0.75
291. LOAD COMB 205 DL+CL+0.75L+0.75TL2
292. 1 1.0 2 1.0 3 0.75 13 0.75
293. LOAD COMB 206 DL+CL+WL1
294. 1 1.0 2 1.0 4 1.0
295. LOAD COMB 207 DL+CL+WL2
296. 1 1.0 2 1.0 5 1.0
297. LOAD COMB 208 DL+CL+WL3
298. 1 1.0 2 1.0 6 1.0
299. LOAD COMB 209 DL+CL+WL4
300. 1 1.0 2 1.0 7 1.0
301. LOAD COMB 210 DL+CL+WL5
302. 1 1.0 2 1.0 8 1.0
303. LOAD COMB 211 DL+CL+WL6
304. 1 1.0 2 1.0 9 1.0
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305. LOAD COMB 212 DL+CL+0.75WL1+0.75L
306. 1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
307. LOAD COMB 213 DL+CL+0.75WL2+0.75L
308. 1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
309. LOAD COMB 214 DL+CL+0.75WL3+0.75L
310. 1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
311. LOAD COMB 215 DL+CL+0.75WL4+0.75L
312. 1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
313. LOAD COMB 216 DL+CL+0.75WL5+0.75L
314. 1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
315. LOAD COMB 217 DL+CL+0.75WL67+0.75L
316. 1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
317. LOAD COMB 218 0.6DL+CL+WL1
318. 1 0.6 2 1.0 4 1.0
319. LOAD COMB 219 0.6DL+CL+WL2
320. 1 0.6 2 1.0 5 1.0
321. LOAD COMB 220 0.6DL+CL+WL3
322. 1 0.6 2 1.0 6 1.0
323. LOAD COMB 221 0.6DL+CL+WL4
324. 1 0.6 2 1.0 7 1.0
325. LOAD COMB 222 0.6DL+CL+WL5
326. 1 0.6 2 1.0 8 1.0
327. LOAD COMB 223 0.6DL+CL+WL6
328. 1 0.6 2 1.0 9 1.0
329. LOAD COMB 224 0.6DL+CL+0.7SL1
330. 1 0.6 2 1.0 10 0.7
331. LOAD COMB 225 0.6DL+CL+0.7SL2
332. 1 0.6 2 1.0 11 0.7
333. ********************************************************************************
334. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
335. LOAD COMB 101 1.4DL+1.4CL
336. 1 1.4 2 1.4
337. LOAD COMB 102 1.2DL+1.2CL+1.2TL1+1.6L
338. 1 1.2 2 1.2 3 1.6 12 1.2
339. LOAD COMB 103 1.2DL+1.2CL+1.2TL2+1.6L
340. 1 1.2 2 1.2 3 1.6 13 1.2
341. LOAD COMB 104 1.2DL+1.2CL+0.5L
342. 1 1.2 2 1.2 3 0.5
343. LOAD COMB 105 1.2DL+1.2CL+1.6WL1+0.5L
344. 1 1.2 2 1.2 4 1.6 3 0.5
345. LOAD COMB 106 1.2DL+1.2CL+1.6WL2+0.5L
346. 1 1.2 2 1.2 5 1.6 3 0.5
347. LOAD COMB 107 1.2DL+1.2CL+1.6WL3+0.5L
348. 1 1.2 2 1.2 6 1.6 3 0.5
349. LOAD COMB 108 1.2DL+1.2CL+1.6WL4+0.5L
350. 1 1.2 2 1.2 7 1.6 3 0.5
351. LOAD COMB 109 1.2DL+1.2CL+1.6WL5+0.5L
352. 1 1.2 2 1.2 8 1.6 3 0.5
353. LOAD COMB 110 1.2DL+1.2CL+1.6WL6+0.5L
354. 1 1.2 2 1.2 9 1.6 3 0.5
355. LOAD COMB 111 1.2DL+1.2CL+1.0SL1+0.5L
356. 1 1.2 2 1.2 3 0.5 10 1.0
357. LOAD COMB 112 1.2DL+1.2CL+1.0SL2+0.5L
358. 1 1.2 2 1.2 3 0.5 11 1.0
359. LOAD COMB 113 0.9DL+0.9CL+1.6WL1
360. 1 0.9 2 0.9 4 1.6
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361. LOAD COMB 114 0.9DL+0.9CL+1.6WL2
362. 1 0.9 2 0.9 5 1.6
363. LOAD COMB 115 0.9DL+0.9CL+1.6WL3
364. 1 0.9 2 0.9 6 1.6
365. LOAD COMB 116 0.9DL+0.9CL+1.6WL4
366. 1 0.9 2 0.9 7 1.6
367. LOAD COMB 117 0.9DL+0.9CL+1.6WL5
368. 1 0.9 2 0.9 8 1.6
369. LOAD COMB 118 0.9DL+0.9CL+1.6WL6
370. 1 0.9 2 0.9 9 1.6
371. LOAD COMB 119 0.9DL+0.9CL+1.0SL1
372. 1 0.9 2 0.9 10 1.0
373. LOAD COMB 120 0.9DL+0.9CL+1.0SL2
374. 1 0.9 2 0.9 11 1.0
375. ********************************************************************************
376. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 290 NUMBER OF MEMBERS 1
NUMBER OF PLATES 256 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 289

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 146/ 29/ 90 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 873
TOTAL LOAD COMBINATION CASES = 45 SO FAR.
SIZE OF STIFFNESS MATRIX = 79 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.3/ 144949.5 MB

```
377. LOAD LIST 101 TO 120
378. START CONCRETE DESIGN
```



```
379. CODE ACI
380. TRACK 2 ALL
381. FC 35000 MEMB 185
382. CLB 0.05 MEMB 185
383. CLS 0.05 ALL
384. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 338 341 TO 432
385. MAXMAIN 20 ALL
386. MINMAIN 20 ALL
387. MINSEC 10 ALL
388. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 4985.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 20 MM	2.011	113	STA	0.900

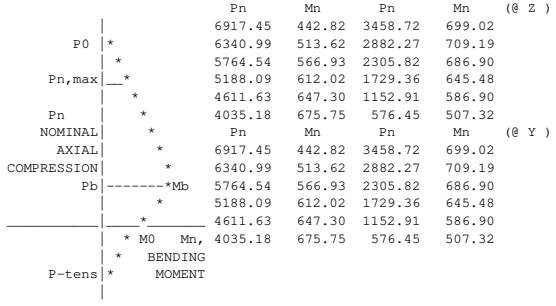
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9367.38	7493.90	3045.15	713.70	234.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
418.53	-2079.42	-195.66	261.76	-0.58168

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
9367.38	7493.90	3045.15	713.70	234.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
418.53	-2079.42	-195.66	163.96	-0.36435



**********END OF COLUMN DESIGN RESULTS**********

389. END CONCRETE DESIGN

390. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 20,2020 TIME= 18:45:52 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUL 20, 2020                *
*           Time=    18:46:24                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1 - STABILITY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/07/2020
    4. JOB NAME FDN. MKD. F1_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 912.2 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 PRIS YD 0.5 ZD 0.5
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FX 10 FY -16.7
   37. ******************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE COLATERAL LOAD ( CL )
```



```
   39. JOINT LOAD
   40. 128 FX 0.1 FY -2.1
   41. ******************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
   43. JOINT LOAD
   44. 128 FX 34.4 FY -46.7
   45. ******************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
   47. JOINT LOAD
   48. 128 FX -69.7 FY 128.72 FZ 40.1
   49. ******************************************************************************
   50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
   51. JOINT LOAD
   52. 128 FX -37 FY 103.62 FZ 40.1
   53. ******************************************************************************
   54. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
   55. JOINT LOAD
   56. 128 FX -50.4 FY 89.52 FZ 40.1
   57. ******************************************************************************
   58. LOAD 7 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
   59. JOINT LOAD
   60. 128 FX -17.6 FY 64.42 FZ 40.1
   61. ******************************************************************************
   62. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
   63. JOINT LOAD
   64. 128 FX -38.7 FY 157.32 FZ 40.1
   65. ******************************************************************************
   66. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
   67. JOINT LOAD
   68. 128 FX -41.7 FY 141.02 FZ 40.1
   69. ******************************************************************************
   70. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
   71. JOINT LOAD
   72. 128 FX 0.5 FY 2.9 FZ 2.8
   73. ******************************************************************************
   74. LOAD 11 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
   75. JOINT LOAD
   76. 128 FX -0.5 FY 2.5 FZ 2.8
   77. ******************************************************************************
   78. LOAD 12 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
   79. JOINT LOAD
   80. 128 FX -0.4
   81. ******************************************************************************
   82. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
   83. JOINT LOAD
   84. 128 FX 0.4
   85. ******************************************************************************
   86. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   87. LOAD COMB 201 DL + CL
   88. 1 1.0 2 1.0
   89. LOAD COMB 202 DL+CL+L+TL1
   90. 1 1.0 2 1.0 3 1.0 12 1.0
   91. LOAD COMB 203 DL+CL+L+TL2
   92. 1 1.0 2 1.0 3 1.0 13 1.0
   93. LOAD COMB 204 DL+CL+0.75L+0.75TL1
   94. 1 1.0 2 1.0 3 0.75 12 0.75
```

```
 95. LOAD COMB 205 DL+CL+0.75L+0.75TL2
 96. 1 1.0 2 1.0 3 0.75 13 0.75
 97. LOAD COMB 206 DL+CL+WL1
 98. 1 1.0 2 1.0 4 1.0
 99. LOAD COMB 207 DL+CL+WL2
100. 1 1.0 2 1.0 5 1.0
101. LOAD COMB 208 DL+CL+WL3
102. 1 1.0 2 1.0 6 1.0
103. LOAD COMB 209 DL+CL+WL4
104. 1 1.0 2 1.0 7 1.0
105. LOAD COMB 210 DL+CL+WL5
106. 1 1.0 2 1.0 8 1.0
107. LOAD COMB 211 DL+CL+WL6
108. 1 1.0 2 1.0 9 1.0
109. LOAD COMB 212 DL+CL+0.75WL1+0.75L
110. 1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
111. LOAD COMB 213 DL+CL+0.75WL2+0.75L
112. 1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
113. LOAD COMB 214 DL+CL+0.75WL3+0.75L
114. 1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
115. LOAD COMB 215 DL+CL+0.75WL4+0.75L
116. 1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
117. LOAD COMB 216 DL+CL+0.75WL5+0.75L
118. 1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
119. LOAD COMB 217 DL+CL+0.75WL67+0.75L
120. 1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
121. LOAD COMB 218 0.6DL+CL+WL1
122. 1 0.6 2 1.0 4 1.0
123. LOAD COMB 219 0.6DL+CL+WL2
124. 1 0.6 2 1.0 5 1.0
125. LOAD COMB 220 0.6DL+CL+WL3
126. 1 0.6 2 1.0 6 1.0
127. LOAD COMB 221 0.6DL+CL+WL4
128. 1 0.6 2 1.0 7 1.0
129. LOAD COMB 222 0.6DL+CL+WL5
130. 1 0.6 2 1.0 8 1.0
131. LOAD COMB 223 0.6DL+CL+WL6
132. 1 0.6 2 1.0 9 1.0
133. LOAD COMB 224 0.6DL+CL+0.7SL1
134. 1 0.6 2 1.0 10 0.7
135. LOAD COMB 225 0.6DL+CL+0.7SL2
136. 1 0.6 2 1.0 11 0.7
137. ********************************************************************************
138. PERFORM ANALYSIS
```


P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**2**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 25 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 144942.9 MB

```
139. FINISH

          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= JUL 20,2020   TIME= 18:46:29 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *              http://www.bentley.com                      *
      ************************************************************
```

ATTACHMENT 5 STAAD OUTPUT – FOUNDATION F2

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    JUL 20, 2020                 *
*          Time=    18:51:21                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04/07/2020
    4. JOB NAME FDN. MKD. F2
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 912.2 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
   20. 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438
   21. 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438
   22. 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438
   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
   28. 344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438
   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
   30. 350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438
   31. 353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438
   32. 356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438
   33. 359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438
   34. 362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438
   35. 365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925
   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
   37. 371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438
   38. 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925
```



```
   39. 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438
   40. 380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925
   41. 383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925
   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
   43. 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438
   44. 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438
   45. 395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438
   46. 398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438
   47. 401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438
   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
   49. 407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438
   50. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   51. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   52. 416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438
   53. 419 -1 909.9 2.92438
   54. MEMBER INCIDENCES
   55. 185 2 128
   56. ELEMENT INCIDENCES SHELL
   57. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
   58. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309
   59. 341 319 320 316 315; 342 320 303 10 316; 343 321 322 302 303
   60. 344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305
   61. 347 325 326 307 306; 348 326 327 308 307; 349 327 328 313 308
   62. 350 328 329 314 313; 351 330 331 320 319; 352 331 321 303 320
   63. 353 332 333 322 321; 354 333 334 323 322; 355 334 335 324 323
   64. 356 335 336 325 324; 357 336 337 326 325; 358 337 338 327 326
   65. 359 338 339 328 327; 360 339 340 329 328; 361 341 342 331 330
   66. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
   67. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
   68. 368 348 349 338 337; 369 349 350 339 338; 370 350 351 340 339
   69. 371 352 353 342 341; 372 353 343 332 342; 373 354 355 344 343
   70. 374 355 356 345 344; 375 356 357 346 345; 376 357 358 347 346
   71. 377 358 359 348 347; 378 359 360 349 348; 379 360 361 350 349
   72. 380 361 362 351 350; 381 363 364 353 352; 382 364 354 343 353
   73. 383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370
   74. 386 371 373 374 372; 387 373 375 376 374; 388 375 377 378 376
   75. 389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386
   76. 392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366
   77. 395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373
   78. 398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379
   79. 401 396 397 384 383; 402 397 387 365 384; 403 398 399 388 387
   80. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
   81. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
   82. 410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397
   83. 413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400
   84. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
   85. 419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407
   86. 422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411
   87. 426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415
   88. 430 309 311 417 416; 431 315 316 419 418; 432 316 10 409 419
   89. START GROUP DEFINITION
   90. ELEMENT
   91. _RAFT 326 TO 328 332 TO 334 337 338 341 TO 432
   92. _RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 432
   93. _RAFT_SHEAR 326 334 337 338 341 TO 343 348 TO 413 418 TO 423 428 TO 432
   94. MEMBER
```

```
 95. END GROUP DEFINITION
 96. ELEMENT PROPERTY
 97. 326 TO 328 332 TO 334 337 338 341 TO 432 THICKNESS 0.5
 98. DEFINE MATERIAL START
 99. ISOTROPIC CONCRETE
100. E 2.17185E+007
101. POISSON 0.17
102. DENSITY 25
103. ALPHA 1E-005
104. DAMP 0.05
105. TYPE CONCRETE
106. STRENGTH FCU 27579
107. END DEFINE MATERIAL
108. MEMBER PROPERTY AMERICAN
109. 185 PRIS YD 0.5 ZD 0.5
110. CONSTANTS
111. MATERIAL CONCRETE ALL
112. SUPPORTS
113. 326 TO 328 332 TO 334 337 338 341 TO 431 -
114. 432 PLATE MAT DIRECT Y SUBGRADE 90000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
10	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
303	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.062500
311	0.031250
313	0.062423
314	0.031211
315	0.031250
316	0.062500
319	0.031211
320	0.062423
321	0.062500
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.062500
328	0.062500
329	0.031250
330	0.031250

331	0.062500
332	0.062500
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.062500
339	0.062500
340	0.031250
341	0.031250
342	0.062500
343	0.062500
344	0.062500
345	0.062500
346	0.062500
347	0.062500
348	0.062500
349	0.062500
350	0.062500
351	0.031250
352	0.031250
353	0.062500
354	0.031250
355	0.031250
356	0.031250
357	0.031250
358	0.031250
359	0.031250
360	0.031250
361	0.031250
362	0.015625
363	0.015625
364	0.031250
365	0.062422
366	0.062422
367	0.031173
368	0.031173
369	0.062422
370	0.031173
371	0.062422
372	0.031173
373	0.062422
374	0.031173
375	0.062422
376	0.031173
377	0.062422
378	0.031173
379	0.062423
380	0.031173
381	0.031211
382	0.015586
383	0.031211
384	0.062423
385	0.031173
386	0.015586

```
       387          0.062500
       388          0.062500
       389          0.062500
       390          0.062500
       391          0.062500
       392          0.062500
       393          0.062500
       394          0.062500
       395          0.031250
       396          0.031250
       397          0.062500
       398          0.062500
       399          0.062500
       400          0.062500
       401          0.062500
       402          0.062500
       403          0.062500
       404          0.062500
       405          0.062500
       406          0.031250
       407          0.031250
       408          0.062500
       409          0.062578
       410          0.062578
       411          0.062578
       412          0.062578
       413          0.062578
       414          0.062578
       415          0.062578
       416          0.062578
       417          0.031289
       418          0.031289
       419          0.062578
    TOTAL           6.248453
115. ********************************************************************************
116. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
117. SELFWEIGHT Y -1
118. ELEMENT LOAD
119. * LOAD DUE TO BACKFILL
120. 326 TO 328 332 TO 334 337 338 341 TO 432 PR GY -20
121. * LOAD DUE TO SURCHARGE
122. 326 TO 328 332 TO 334 337 338 341 TO 432 PR GY -20
123. JOINT LOAD
124. 128 FY -5.9
125. ********************************************************************************
126. LOAD 2 LOADTYPE NONE  TITLE COLATERAL LOAD ( CL )
127. JOINT LOAD
128. 128 FY -0.2
129. ********************************************************************************
130. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
131. JOINT LOAD
132. 128 FY -13.2
133. ********************************************************************************
134. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
135. JOINT LOAD
136. 128 FY 31.3 FZ 22.3
```



```
137. ********************************************************************************
138. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
139. JOINT LOAD
140. 128 FY 28.6 FZ 24.6
141. ********************************************************************************
142. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
143. JOINT LOAD
144. 128 FY 21.5 FZ 24.6
145. ********************************************************************************
146. LOAD 7 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
147. JOINT LOAD
148. 128 FY 18.8 FZ 22.3
149. ********************************************************************************
150. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
151. JOINT LOAD
152. 128 FY 26.8 FZ 22.3
153. ********************************************************************************
154. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
155. JOINT LOAD
156. 128 FY 12.6 FZ 24.6
157. ********************************************************************************
158. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
159. JOINT LOAD
160. 128 FY -0.5
161. ********************************************************************************
162. LOAD 11 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
163. JOINT LOAD
164. 128 FY 0.5
165. ********************************************************************************
166. LOAD 12 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
167. JOINT LOAD
168. 128 FY 2.7
169. ********************************************************************************
170. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
171. JOINT LOAD
172. 128 FY -2.7
173. ********************************************************************************
174. * UNFACTORED LOAD COMBINATIONS
175. LOAD COMB 201 DL + CL
176. 1 1.0 2 1.0
177. LOAD COMB 202 DL+CL+L+TL1
178. 1 1.0 2 1.0 3 1.0 12 1.0
179. LOAD COMB 203 DL+CL+L+TL2
180. 1 1.0 2 1.0 3 1.0 13 1.0
181. LOAD COMB 204 DL+CL+0.75L+0.75TL1
182. 1 1.0 2 1.0 3 0.75 12 0.75
183. LOAD COMB 205 DL+CL+0.75L+0.75TL2
184. 1 1.0 2 1.0 3 0.75 13 0.75
185. LOAD COMB 206 DL+CL+WL1
186. 1 1.0 2 1.0 4 1.0
187. LOAD COMB 207 DL+CL+WL2
188. 1 1.0 2 1.0 5 1.0
189. LOAD COMB 208 DL+CL+WL3
190. 1 1.0 2 1.0 6 1.0
191. LOAD COMB 209 DL+CL+WL4
192. 1 1.0 2 1.0 7 1.0
```

```
193. LOAD COMB 210 DL+CL+WL5
194. 1 1.0 2 1.0 8 1.0
195. LOAD COMB 211 DL+CL+WL6
196. 1 1.0 2 1.0 9 1.0
197. LOAD COMB 212 DL+CL+0.75WL1+0.75L
198. 1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
199. LOAD COMB 213 DL+CL+0.75WL2+0.75L
200. 1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
201. LOAD COMB 214 DL+CL+0.75WL3+0.75L
202. 1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
203. LOAD COMB 215 DL+CL+0.75WL4+0.75L
204. 1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
205. LOAD COMB 216 DL+CL+0.75WL5+0.75L
206. 1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
207. LOAD COMB 217 DL+CL+0.75WL67+0.75L
208. 1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
209. LOAD COMB 218 0.6DL+CL+WL1
210. 1 0.6 2 1.0 4 1.0
211. LOAD COMB 219 0.6DL+CL+WL2
212. 1 0.6 2 1.0 5 1.0
213. LOAD COMB 220 0.6DL+CL+WL3
214. 1 0.6 2 1.0 6 1.0
215. LOAD COMB 221 0.6DL+CL+WL4
216. 1 0.6 2 1.0 7 1.0
217. LOAD COMB 222 0.6DL+CL+WL5
218. 1 0.6 2 1.0 8 1.0
219. LOAD COMB 223 0.6DL+CL+WL6
220. 1 0.6 2 1.0 9 1.0
221. LOAD COMB 224 0.6DL+CL+0.7SL1
222. 1 0.6 2 1.0 10 0.7
223. LOAD COMB 225 0.6DL+CL+0.7SL2
224. 1 0.6 2 1.0 11 0.7
225. ********************************************************************************
226. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
227. LOAD COMB 101 1.4DL+1.4CL
228. 1 1.4 2 1.4
229. LOAD COMB 102 1.2DL+1.2CL+1.2TL1+1.6L
230. 1 1.2 2 1.2 3 1.6 12 1.2
231. LOAD COMB 103 1.2DL+1.2CL+1.2TL2+1.6L
232. 1 1.2 2 1.2 3 1.6 13 1.2
233. LOAD COMB 104 1.2DL+1.2CL+0.5L
234. 1 1.2 2 1.2 3 0.5
235. LOAD COMB 105 1.2DL+1.2CL+1.6WL1+0.5L
236. 1 1.2 2 1.2 4 1.6 3 0.5
237. LOAD COMB 106 1.2DL+1.2CL+1.6WL2+0.5L
238. 1 1.2 2 1.2 5 1.6 3 0.5
239. LOAD COMB 107 1.2DL+1.2CL+1.6WL3+0.5L
240. 1 1.2 2 1.2 6 1.6 3 0.5
241. LOAD COMB 108 1.2DL+1.2CL+1.6WL4+0.5L
242. 1 1.2 2 1.2 7 1.6 3 0.5
243. LOAD COMB 109 1.2DL+1.2CL+1.6WL5+0.5L
244. 1 1.2 2 1.2 8 1.6 3 0.5
245. LOAD COMB 110 1.2DL+1.2CL+1.6WL6+0.5L
246. 1 1.2 2 1.2 9 1.6 3 0.5
247. LOAD COMB 111 1.2DL+1.2CL+1.0SL1+0.5L
248. 1 1.2 2 1.2 3 0.5 10 1.0
```



```
249. LOAD COMB 112 1.2DL+1.2CL+1.0SL2+0.5L
250. 1 1.2 2 1.2 3 0.5 11 1.0
251. LOAD COMB 113 0.9DL+0.9CL+1.6WL1
252. 1 0.9 2 0.9 4 1.6
253. LOAD COMB 114 0.9DL+0.9CL+1.6WL2
254. 1 0.9 2 0.9 5 1.6
255. LOAD COMB 115 0.9DL+0.9CL+1.6WL3
256. 1 0.9 2 0.9 6 1.6
257. LOAD COMB 116 0.9DL+0.9CL+1.6WL4
258. 1 0.9 2 0.9 7 1.6
259. LOAD COMB 117 0.9DL+0.9CL+1.6WL5
260. 1 0.9 2 0.9 8 1.6
261. LOAD COMB 118 0.9DL+0.9CL+1.6WL6
262. 1 0.9 2 0.9 9 1.6
263. LOAD COMB 119 0.9DL+0.9CL+1.0SL1
264. 1 0.9 2 0.9 10 1.0
265. LOAD COMB 120 0.9DL+0.9CL+1.0SL2
266. 1 0.9 2 0.9 11 1.0
267. ********************************************************************************
268. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**122**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**100**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**121**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 119/ 20/ 66 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 369
TOTAL LOAD COMBINATION CASES = 45 SO FAR.
SIZE OF STIFFNESS MATRIX = 25 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.8/ 144942.8 MB

```
269. LOAD LIST 101 TO 120
270. START CONCRETE DESIGN
```

```
271. CODE ACI
272. TRACK 2 ALL
273. FC 35000 MEMB 185
274. CLB 0.05 MEMB 185
275. CLS 0.05 ALL
276. CLT 0.05 ALL
277. MAXMAIN 20 ALL
278. MINMAIN 20 ALL
279. MINSEC 10 ALL
280. DESIGN COLUMN 185
```



```
================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 500.0 X 500.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  2500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 20 MM               1.005      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 8402.44   6721.95   3052.41    594.14     194.6
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 222.73  -1039.71     13.14      0.00    0.00000
 ------------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 8402.44   6721.95   3052.41    594.14     194.6
   M0        P-tens.   Des.Pn    Des.Mn     e/h
 222.73  -1039.71     13.14      0.00    0.00000
 ------------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              6204.88   403.06  3102.44   593.55
       P0 |*             5687.81   461.79  2585.37   569.74
          | *            5170.73   509.07  2068.29   531.80
    Pn,max|__*           4653.66   544.26  1551.22   482.01
          |   *          4136.59   568.89  1034.15   410.92
    Pn    |    *         3619.51   584.62   517.07   321.89
   NOMINAL|     *          Pn       Mn       Pn       Mn    (@ Y )
     AXIAL|      *       6204.88   403.06  3102.44   593.55
COMPRESSION|       *      5687.81   461.79  2585.37   569.74
         Pb|-------*Mb    5170.73   509.07  2068.29   531.80
           |      *       4653.66   544.26  1551.22   482.01
___________|____*_______  4136.59   568.89  1034.15   410.92
           |  * M0   Mn,  3619.51   584.62   517.07   321.89
           | *   BENDING
    P-tens |*     MOMENT
           |


********************END OF COLUMN DESIGN RESULTS********************
```

```
281. END CONCRETE DESIGN
282. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= JUL 20,2020   TIME= 18:51:29 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
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```

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            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    JUL 20, 2020                *
            *           Time=    18:52:32                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2 - STABILITY CHECK.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 04/07/2020
     4. JOB NAME FDN. MKD. F2_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 909.9 3.175; 128 0 912.2 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 0.5 ZD 0.5
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 2 FIXED
    32. ******************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 128 FY -5.9
    37. ******************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE COLATERAL LOAD ( CL )
```



```
    39. JOINT LOAD
    40. 128 FY -0.2
    41. ******************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE LIVE LOAD ( L )
    43. JOINT LOAD
    44. 128 FY -13.2
    45. ******************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD  ( WL1 )
    47. JOINT LOAD
    48. 128 FY 31.3 FZ 22.3
    49. ******************************************************************************
    50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD  ( WL2 )
    51. JOINT LOAD
    52. 128 FY 28.6 FZ 24.6
    53. ******************************************************************************
    54. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD  ( WL3 )
    55. JOINT LOAD
    56. 128 FY 21.5 FZ 24.6
    57. ******************************************************************************
    58. LOAD 7 LOADTYPE NONE  TITLE WIND LOAD  ( WL4 )
    59. JOINT LOAD
    60. 128 FY 18.8 FZ 22.3
    61. ******************************************************************************
    62. LOAD 8 LOADTYPE NONE  TITLE WIND LOAD  ( WL5 )
    63. JOINT LOAD
    64. 128 FY 26.8 FZ 22.3
    65. ******************************************************************************
    66. LOAD 9 LOADTYPE NONE  TITLE WIND LOAD  ( WL6 )
    67. JOINT LOAD
    68. 128 FY 12.6 FZ 24.6
    69. ******************************************************************************
    70. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL1 )
    71. JOINT LOAD
    72. 128 FY -0.5
    73. ******************************************************************************
    74. LOAD 11 LOADTYPE NONE  TITLE SEISMIC LOAD  ( SL2 )
    75. JOINT LOAD
    76. 128 FY 0.5
    77. ******************************************************************************
    78. LOAD 12 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL1 )
    79. JOINT LOAD
    80. 128 FY 2.7
    81. ******************************************************************************
    82. LOAD 13 LOADTYPE NONE  TITLE TEMPERATURE LOAD  ( TL2 )
    83. JOINT LOAD
    84. 128 FY -2.7
    85. ******************************************************************************
    86. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
    87. LOAD COMB 201 DL + CL
    88. 1 1.0 2 1.0
    89. LOAD COMB 202 DL+CL+L+TL1
    90. 1 1.0 2 1.0 3 1.0 12 1.0
    91. LOAD COMB 203 DL+CL+L+TL2
    92. 1 1.0 2 1.0 3 1.0 13 1.0
    93. LOAD COMB 204 DL+CL+0.75L+0.75TL1
    94. 1 1.0 2 1.0 3 0.75 12 0.75
```

```
 95. LOAD COMB 205 DL+CL+0.75L+0.75TL2
 96. 1 1.0 2 1.0 3 0.75 13 0.75
 97. LOAD COMB 206 DL+CL+WL1
 98. 1 1.0 2 1.0 4 1.0
 99. LOAD COMB 207 DL+CL+WL2
100. 1 1.0 2 1.0 5 1.0
101. LOAD COMB 208 DL+CL+WL3
102. 1 1.0 2 1.0 6 1.0
103. LOAD COMB 209 DL+CL+WL4
104. 1 1.0 2 1.0 7 1.0
105. LOAD COMB 210 DL+CL+WL5
106. 1 1.0 2 1.0 8 1.0
107. LOAD COMB 211 DL+CL+WL6
108. 1 1.0 2 1.0 9 1.0
109. LOAD COMB 212 DL+CL+0.75WL1+0.75L
110. 1 1.0 2 1.0 4 0.75 3 0.75 3 0.75
111. LOAD COMB 213 DL+CL+0.75WL2+0.75L
112. 1 1.0 2 1.0 5 0.75 3 0.75 3 0.75
113. LOAD COMB 214 DL+CL+0.75WL3+0.75L
114. 1 1.0 2 1.0 6 0.75 3 0.75 3 0.75
115. LOAD COMB 215 DL+CL+0.75WL4+0.75L
116. 1 1.0 2 1.0 7 0.75 3 0.75 3 0.75
117. LOAD COMB 216 DL+CL+0.75WL5+0.75L
118. 1 1.0 2 1.0 8 0.75 3 0.75 3 0.75
119. LOAD COMB 217 DL+CL+0.75WL67+0.75L
120. 1 1.0 2 1.0 9 0.75 3 0.75 3 0.75
121. LOAD COMB 218 0.6DL+CL+WL1
122. 1 0.6 2 1.0 4 1.0
123. LOAD COMB 219 0.6DL+CL+WL2
124. 1 0.6 2 1.0 5 1.0
125. LOAD COMB 220 0.6DL+CL+WL3
126. 1 0.6 2 1.0 6 1.0
127. LOAD COMB 221 0.6DL+CL+WL4
128. 1 0.6 2 1.0 7 1.0
129. LOAD COMB 222 0.6DL+CL+WL5
130. 1 0.6 2 1.0 8 1.0
131. LOAD COMB 223 0.6DL+CL+WL6
132. 1 0.6 2 1.0 9 1.0
133. LOAD COMB 224 0.6DL+CL+0.7SL1
134. 1 0.6 2 1.0 10 0.7
135. LOAD COMB 225 0.6DL+CL+0.7SL2
136. 1 0.6 2 1.0 11 0.7
137. ********************************************************************************
138. PERFORM ANALYSIS
```


P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 13, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 25 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 144940.0 MB

```
139. FINISH

          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= JUL 20,2020   TIME= 18:52:37 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
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```